5G RAN

CoMP Feature Parameter Description

Issue Draft B

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
None	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Revised descriptions in the document.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 07 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationships between intra-base-station UL CoMP/intra-base-station DL CoMP and time-domain power saving BWP (low-frequency TDD). For details, see: • **4.3.2.3 Function Impacts** • **5.3.2.3 Function Impacts**	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationship between L/NR Inter-band Deep Coordination and intra-base-station DL CoMP. For details, see **5.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between uplink 4-layer transmission for a single FWA user and intra-base-station UL CoMP. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added support for optimized cooperating cell selection. For details, see: • 4.1.1 Joint Reception by Common Cells • 4.3.2.1 Prerequisite Functions • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands • 5.1.1 Joint Transmission by Common Cells • 5.3.2.1 Prerequisite Functions • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameter: Added the **COOP_CELL_ALLOC_OPT_SW** option to the **NRDUCellComp**.***CompOptSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for intra-base-station UL CoMP for UEs with specific QCIs. For details, see: • 4.1.1 Joint Reception by Common Cells • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added the **NRDUCellComp**.***CompConfigPol*** parameter. Modified parameter: Added the **COMP_SW** option to the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationships of "uplink channel densified equalization for high-speed UEs" with intra-base-station UL CoMP and intra-base-station DL CoMP. For details, see 4.3.2.3 Function Impacts and 5.3.2.3 Function Impacts.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Revised intra-base-station DL CoMP configurations for distributed massive MIMO cells. For details, see **5.1.4 Joint Transmission by Distributed Massive MIMO Cells (NR TDD)**.	None	Low-frequency TDD	DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Revised other descriptions in the document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-030204	CoMP	4 Intra-Base-Station UL CoMP 5 Intra-Base-Station DL CoMP

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Intra-base-station UL CoMP	None	4 Intra-Base-Station UL CoMP
Intra-base-station DL CoMP	None	5 Intra-Base-Station DL CoMP

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Intra-base-station UL CoMP	Supported only in low frequency bands	4 Intra-Base-Station UL CoMP
Intra-base-station DL CoMP	Supported only in low frequency bands	5 Intra-Base-Station DL CoMP

3 Overview

Intra-frequency networking is generally used for NR networks to improve spectrum resource utilization. However, user experience of cell edge users (CEUs) is poor in intra-frequency networking due to high inter-cell interference. Coordinated multipoint transmission/reception (CoMP) can be used to improve user experience of such UEs.

CoMP enables coordinated processing of a UE's PUSCH/PDSCH data by different transmission points to maximize use of time-, frequency-, or space-domain resources of these transmission points. This avoids interference between transmission points or converts it into signals useful for UEs, thereby improving user experience of CEUs.

Based on the transmission node, CoMP involves coordinated processing between different cells and between different transmission reception points (TRPs) in one cell. Currently, coordination is supported only between cells or TRPs served by the same gNodeB. A TRP is defined as an antenna array with one or more antenna elements available to the network. One or more TRPs can be contained in a cell.

Based on the data transmission direction, CoMP is classified into intra-base-station UL CoMP and intra-base-station DL CoMP.

- Intra-base-station UL CoMP enables cells/TRPs under the same gNodeB to jointly receive data from one UE. For details, see **4 Intra-Base-Station UL CoMP**.
- Intra-base-station DL CoMP enables cells/TRPs under the same gNodeB to jointly transmit data to one UE. For details, see **5 Intra-Base-Station DL CoMP**.

For ease of description, this function will be described on a per TRP basis in the context of combined cell or hyper cell scenarios where one cell includes multiple TRPs, and on a per cell basis in common cell scenarios where one cell corresponds to one TRP. **Table 3-1** describes the concepts related to CoMP in common cells.

Table 3-1 Related concepts

Name	Definition
Serving cell	Cell where a UE accesses

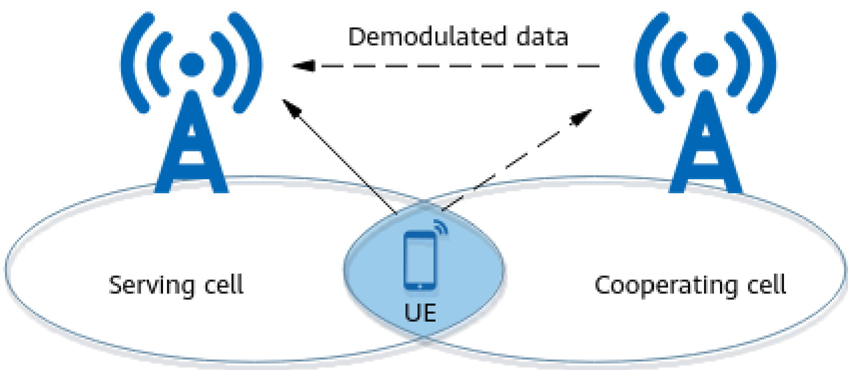
Name	Definition
Cooperating cell	An intra-frequency neighboring cell that works with the serving cell for joint transmission or reception
Overlapping area	Coverage overlap between cells
CoMP UE	A UE for which the serving and cooperating cells perform joint reception or transmission

4 Intra-Base-Station UL CoMP

4.1 Principles

Intra-base-station UL CoMP allows a gNodeB to use antennas of the serving cell and intra-base-station intra-frequency neighboring cells to jointly receive PUSCH signals from a UE in the overlapping area and then combine these signals in the serving cell. Figure 4-1 shows how this function works.

Figure 4-1 Intra-base-station UL CoMP



Note: The serving and cooperating cells are served by the same gNodeB.

This function is controlled by the **INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter. For each UE, its serving cell can jointly receive data with only one cooperating cell.

Intra-base-station UL CoMP involves scenarios listed in **Table 4-1**.

Table 4-1 Scenarios involved in intra-base-station UL CoMP

Scenario	Description	Supported RAT	Chapter/Section
By common cells	Both the serving and cooperating cells of a UE are common cells.	Low-frequency TDD	**4.1.1 Joint Reception by Common Cells**
By TRPs in a combined cell or hyper cell	A UE is served by a TRP in a combined cell or hyper cell, and the gNodeB selects another TRP in the cell for joint data reception.	Low-frequency TDD	**4.1.2 Joint Reception by TRPs in a Combined Cell or Hyper Cell**
By a combined cell and a common cell	A UE is served by a combined cell, and the gNodeB selects an intra-base-station common cell as a cooperating cell for joint data reception.	Low-frequency TDD	**4.1.3 Joint Reception by a Combined Cell and a Common Cell (NR TDD)**
By a combined cell in joint transmission mode and a common cell	A UE is served by a combined cell in joint transmission mode, and the gNodeB selects an intra-base-station common cell as a cooperating cell for joint data reception.	Low-frequency TDD	**4.1.4 Joint Reception by a Combined Cell in Joint Transmission Mode and a Common Cell**

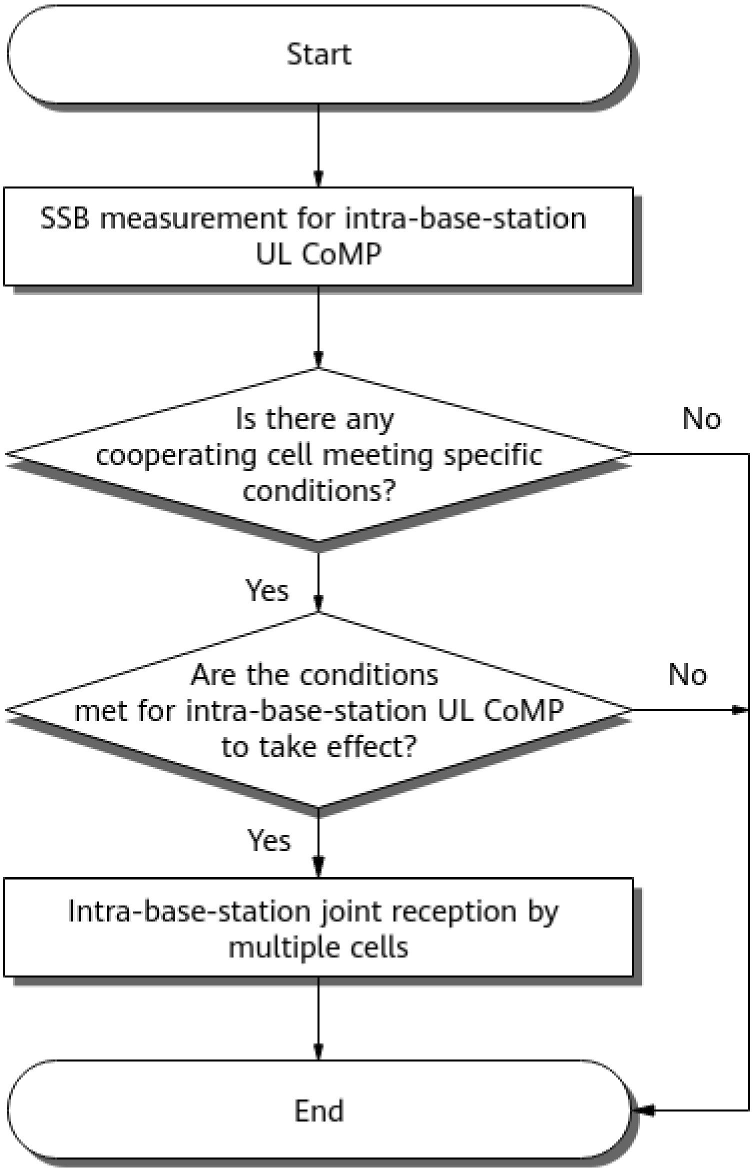
NOTE

- Common cell: **NRDUCell**.***NrDuCellNetworkingMode*** is set to **NORMAL_CELL**.
- Combined cell: **NRDUCell**.***NrDuCellNetworkingMode*** is set to **HYPER_CELL_COMBINE_MODE**, and **NRDUCellMultiTrp**.***DataTransMode*** is set to **BASIC_MODE** or **DPS_MODE**.
- Combined cell in joint transmission mode: **NRDUCell**.***NrDuCellNetworkingMode*** is set to **HYPER_CELL_COMBINE_MODE**, and **NRDUCellMultiTrp**.***DataTransMode*** is set to **JOINT_MODE**.
- Hyper cell: **NRDUCell**.***NrDuCellNetworkingMode*** is set to **HYPER_CELL**.
- For details about common cells and TRPs, see *Cell Management*.
- For details about combined cells in joint transmission mode and combined cells, see *Cell Combination*.
- For details about hyper cells, see *Hyper Cell*.

4.1.1 Joint Reception by Common Cells

When a UE is served by a common cell, an intra-base-station common cell can be selected as a cooperating cell for joint reception. **Figure 4-2** shows the joint reception procedure.

Figure 4-2 Joint reception by common cells



1. Performing SSB measurement for intra-base-station UL CoMP

   The gNodeB selects UEs for synchronization signal and PBCH block (SSB) RSRP measurement as follows:

   – If the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp**.***CompConfigPol*** parameter is deselected, the gNodeB delivers measurement configurations to all UEs in the cell.

   – If the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp**.***CompConfigPol*** parameter is selected, the gNodeB delivers measurement configurations to specific UEs in the cell. Specific UEs are those having at least one QCI for which the **COMP_SW** option of the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected.

   After receiving the measurement configuration, a UE starts SSB RSRP measurement and sends an event A3 measurement report when a certain condition is met.

Event A3 indicates that the signal quality of a neighboring cell is higher than that of the serving cell by a certain offset for a period of time. A UE reports event A3 if the signal quality meets the condition (*Mn* + *Ofn* + *Ocn* – *Hys* > *Ms* + *Ofs* + *Ocs* + *Off*) throughout *TimeToTrigger*. **Table 4-2** explains the variables and lists the parameters used to control them.

Table 4-2 Meanings of variables in event A3 and parameters used to control them

Variable	Meaning	Parameter
TimeToTrigger	Period throughout which the entering condition must be met before the event is reported	**NRCellComp**.***TimeToTrigger***
Ms and *Mn*	Measurement results of the serving and neighboring cells, respectively	None
Ofs and *Ofn*	Frequency-specific offsets of the serving and neighboring cells, respectively. The frequency-specific offset is 0 for intra-frequency cells.	None
Ocs and *Ocn*	Cell individual offsets (CIOs) of the serving and neighboring cells, respectively	The value of *Ocs* is 0, and *Ocn* is specified by the **NRCellRelation**.***CellIndividualOffset*** parameter.
Hys	Hysteresis for the measurement result	**NRCellComp**.***Hysteresis***
Off	Offset for the measurement result	**NRCellComp**.***UlCompRsrpOffset***

For more details about event A3, see section 5.5 "Measurements" in 3GPP TS 38.331 V15.5.0.

2. Determining whether any cooperating cell meets specific conditions

   The gNodeB selects intra-base-station neighboring cells that meet all of the following conditions from the event A3 measurement report:

   – Intra-base-station UL CoMP is enabled in the cells.
   – The cells are configured as intra-frequency neighboring cells of the serving cell.
   – The number of available uplink RBs for the cells is the same as that for the serving cell.
   – In TDD, the TX/RX modes of the cells meet the requirements described in **Table 4-3**.

Table 4-3 TX/RX modes of the serving and cooperating cells required by intra-base-station UL CoMP in TDD common cells

Serving Cell	Cooperating Cell
1T1R	1T1R
2T2R	2T2R, 8T8R, 32T32R, or 64T64R
4T4R	4T4R, 8T8R, 32T32R, or 64T64R
8T8R, 32T32R, or 64T64R	2T2R, 4T4R, 8T8R, 32T32R, or 64T64R

– The parameters listed in **Table 4-4** of the cells have the same settings as those of the serving cell.

Table 4-4 Parameters whose settings must be the same in the serving and cooperating cells

Parameter Name	Parameter	Description
Duplex Mode	**NRDUCell**.***DuplexMode***	None
Downlink NARFCN	**NRDUCell**.***DlNarfcn***	None
Uplink Bandwidth	**NRDUCell**.***UlBandwidth***	None
Slot Assignment	**NRDUCell**.***SlotAssignment***	Applicable only to TDD
Cyclic Prefix Length	**NRDUCell**.***CyclicPrefixLength***	None
Subcarrier Spacing	**NRDUCell**.***SubcarrierSpacing***	None
Uplink DMRS Type	**NRDUCellPusch**.***UlDmrsType***	None
Uplink Additional DMRS Position	**NRDUCellPusch**.***UlAdditionalDmrsPos***	None

If one or more neighboring cells meet all the preceding conditions, the neighboring cell with the optimal signal quality is treated as a cooperating cell for intra-base-station UL CoMP and the UE that sends the event A3 measurement report is treated as an intra-base-station UL CoMP UE. Otherwise, the procedure ends.

In scenarios with dense deployment of multiple cells, intra-base-station UL CoMP may not take effect due to failure to select a suitable cooperating cell. To address this issue, you are advised to enable optimized cooperating cell selection (by selecting the **COOP_CELL_ALLOC_OPT_SW** option of the **NRDUCellComp**.***CompOptSwitch*** parameter) to expand the range of cells that can be selected as cooperating cells in overlapping areas, thereby increasing the probability of intra-base-station UL CoMP taking effect.

3. Determining whether the conditions for intra-base-station UL CoMP to take effect are met

The gNodeB measures the SRS RSRP values of the UL CoMP UE in the serving and cooperating cells. If the SRS RSRP in the cooperating cell minus that in the serving cell is greater than the value of **NRCellComp**.***UlCompRsrpOffset***, joint reception by multiple cells starts. Otherwise, intra-base-station UL CoMP cannot take effect.

4. Performing intra-base-station joint reception by multiple cells

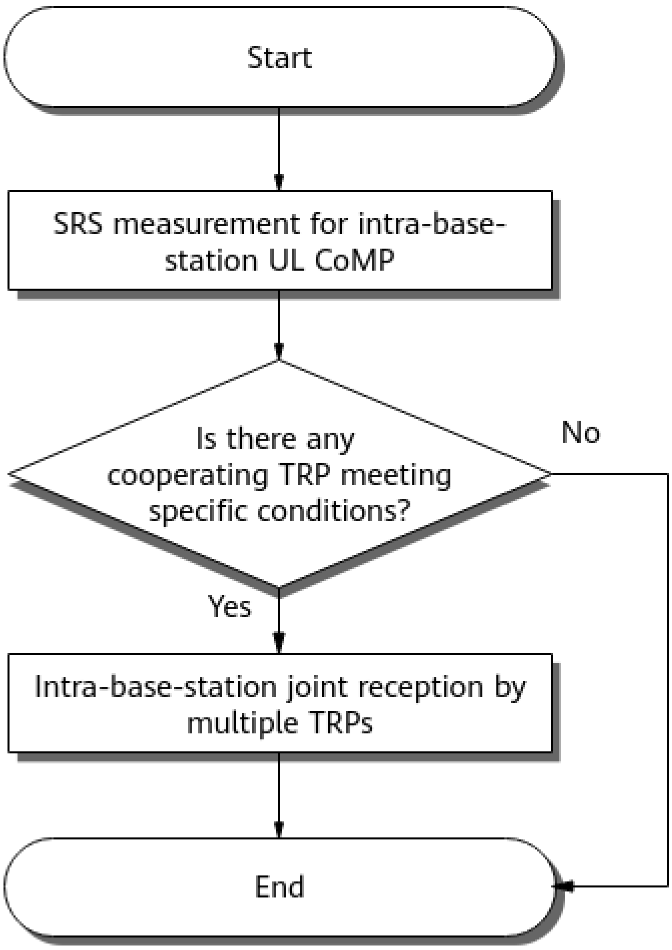
   The serving and cooperating cells jointly receive PUSCH data from the UE. The cooperating cell sends the demodulated data to the serving cell, which then performs combining and decoding.

   The cooperating cell preferentially processes the data of UEs it serves. It then uses its remaining capacity to process the data of the UE involved in joint reception.

4.1.2 Joint Reception by TRPs in a Combined Cell or Hyper Cell

If a UE is served by a combined cell or a hyper cell, other TRPs in the combined cell or hyper cell can be selected for joint reception. **Figure 4-3** shows the joint reception procedure.

Figure 4-3 Joint reception by TRPs in a combined cell or hyper cell



1. Performing SRS measurement for intra-base-station UL CoMP

The gNodeB measures the UE SRS RSRP at each TRP in the combined cell or hyper cell.

2. Determining whether any cooperating TRP meets specific conditions

   If some TRPs meet all of the following conditions, the TRP with the optimal signal quality is treated as a cooperating TRP for intra-base-station UL CoMP.

   – Low-speed scenarios (**NRDUCell**.***HighSpeedFlag*** set to **LOW_SPEED**)

     ▪ In TDD, the TX/RX modes of the TRPs meet the requirements described in **Table 4-5**.

Table 4-5 TX/RX modes of the serving and cooperating TRPs required by intra-base-station UL CoMP in a low-speed TDD combined cell or hyper cell

Serving TRP	Cooperating TRP
1T1R	1T1R
2T2R	2T2R or 8T8R
4T4R	4T4R or 8T8R
8T8R	2T2R, 4T4R, or 8T8R
32T32R or 64T64R	32T32R or 64T64R
Note: Only hyper cells support 32T32R or 64T64R.	

   – High-speed scenarios (**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**)

     ▪ In TDD, the TX/RX modes of the TRPs meet the requirements described in **Table 4-6**.

Table 4-6 TX/RX modes of the serving and cooperating TRPs required by intra-base-station UL CoMP in a high-speed TDD combined cell or hyper cell

Serving TRP	Cooperating TRP
1T1R	1T1R
4T4R	4T4R
2T2R or 8T8R	2T2R or 8T8R

   – The signal quality meets the following requirements: Cooperating TRP RSRP – **NRCellComp**.***Hysteresis*** – Serving TRP RSRP > **NRCellComp**.***UlCompRsrpOffset***.

   If no TRP meets the preceding conditions, the procedure ends.

3. Performing intra-base-station joint reception by multiple TRPs

   The cooperating TRP sends the demodulated data to the serving TRP, which then performs combining and decoding. The cooperating TRP preferentially processes the data of UEs it serves. It then uses its remaining capacity to process the data of the UE involved in joint reception.

4.1.3 Joint Reception by a Combined Cell and a Common Cell (NR TDD)

When a UE is served by a combined cell, an intra-base-station common cell can be selected as a cooperating cell for joint reception if the TRPs of the combined cell work in the same TX/RX mode (2T2R, 4T4R, or 8T8R) and the cooperating cell works in 32T32R or 64T64R. The processing procedure is the same as that for joint reception by common cells. For details, see **4.1.1 Joint Reception by Common Cells**.

4.1.4 Joint Reception by a Combined Cell in Joint Transmission Mode and a Common Cell

In low-speed scenarios, when a UE is served by a combined cell in joint transmission mode, an intra-base-station common cell can be selected as a cooperating cell for joint reception. The processing procedure is the same as that for joint reception by common cells. For details, see **4.1.1 Joint Reception by Common Cells**.

In TDD, all TRPs in a combined cell in joint transmission mode must work in 2T2R mode, and the cooperating cell must work in 32T32R or 64T64R mode.

4.2 Network Analysis

4.2.1 Benefits

In intra-frequency continuous networking, enabling UL CoMP increases the average uplink UE throughput in the overlapping area between intra-base-station cells. The specific gain depends on factors such as the SRS RSRP difference (cooperating cell SRS RSRP – serving cell SRS RSRP), traffic model, network load, and interference. The gain is higher in the case of a greater RSRP difference, larger data packets, or lighter network load. In the case of strong interference, the gain decreases.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

Intra-base-station UL CoMP improves the demodulation performance and spectral efficiency of UL CoMP UEs and slightly decreases the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) of the cells.

When intra-base-station UL CoMP takes effect during a handover, the following changes may be observed in drive tests: increased handover delay, decreased SSB RSRP, and fluctuation in the values of indicators such as the IBLER/RBLER and average throughput in both the uplink and downlink.

After intra-base-station UL CoMP is enabled, the average uplink rank of the cell increases as the proportion of uplink rank 2 increases for 2T UEs. When the SINR differs greatly between two layers for a UE, the spectral efficiency calculated based on the average SINR of the two layers is not optimal. In this case, it is recommended that the **UL_ADAPTIVE_RANK_DEC_SW** option of the **NRDUCellUlRank**.***UlRankAlgoSw*** parameter be selected to ensure more accurate rank selection, thereby increasing the average uplink UE throughput and average uplink cell throughput.

Table 4-7 Definition of related indicators

Indicator	Formula
Uplink PRB usage	**Uplink Resource Block Utilizing Rate (DU)**
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
Downlink IBLER	(**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100%
Uplink RBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
Downlink RBLER	(**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100%

Indicator	Formula
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Average downlink cell throughput	**Cell Downlink Average Throughput (DU)**
Average uplink UE throughput	**User Uplink Average Throughput (DU)**
Average downlink UE throughput	**User Downlink Average Throughput (DU)**
Average uplink rank	SUM(***k* x N.PUSCH.InitTbUl.Rank.*k***)/SUM(**N.PUSCH.InitTbUl.Rank.*k***) *k* = 1 to 2

4.3 Requirements

4.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
Low-frequency TDD	FOFD-030204	CoMP	NR0S000CMP00	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Slot configuration	**NRDUCell**.***SlotAssignment***	*Standards Compliance*	When intra-base-station UL CoMP is enabled, the **NRDUCell**.***SlotAssignment*** parameter can only be set to **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**.
Low-frequency TDD	SRS interference coordination based on self-contained slots	**SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Channel Management*	In low-frequency NR TDD scenarios, optimized cooperating cell selection (controlled by the **COOP_CELL_ALLOC_OPT_SW** option of the **NRDUCellComp**.***CompOptSwitch*** parameter) requires that SRS interference coordination based on self-contained slots be enabled first.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	Intra-base-station UL CoMP cannot be enabled for a high-speed cell if the cell is a common cell (with **NRDUCell**.***NrDuCellNetworkingMode*** set to **NORMAL_CELL**).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	**DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter	*Distributed Antenna Solutions (Low-Frequency TDD)*	Intra-base-station UL CoMP is not supported between distributed massive MIMO cells.
Low-frequency TDD	U-TDOA-based positioning/TOA fingerprint-based positioning	**UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	U-TDOA-based positioning/TOA fingerprint-based positioning cannot be enabled together with intra-base-station UL CoMP.
Low-frequency TDD	SRS field strength-based positioning	**SRS_RSRP_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	SRS field strength-based positioning cannot be enabled together with intra-base-station UL CoMP.
Low-frequency TDD	Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	Intra-base-station UL CoMP is not supported in fusion cells.

4.3.2.3 Function Impacts

- Functions related to carrier management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When carrier aggregation (CA) takes effect, intra-base-station UL CoMP cannot be performed for CA UEs in their SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When CA takes effect, intra-base-station UL CoMP cannot be performed for CA UEs in their SCells.

- Functions related to spectrum management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RAN sharing with common carrier	**gNBSharingMode**.***gNBMultiOpSharingMode*** set to **SHARED_FREQ**	*Multi-Operator Sharing*	The PLMN list of a cooperating cell of a UE must contain the ID of the PLMN with which the UE has registered.
Low-frequency TDD	RAN sharing with dedicated carrier	**gNBSharingMode**.***gNBMultiOpSharingMode*** set to **SEPARATED_FREQ**	*Multi-Operator Sharing*	The PLMN list of a cooperating cell of a UE must contain the ID of the PLMN with which the UE has registered.
Low-frequency TDD	LTE TDD and NR spectrum sharing	**LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	None	When LTE TDD and NR spectrum sharing takes effect, the number of available downlink RBs in the NR cell dynamically changes. This may affect the probability that intra-base-station UL CoMP takes effect.

- Functions related to cell networking planning

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Intra-base-station UL CoMP can be performed between TRPs in a hyper cell; but it cannot be performed between a hyper cell and a common cell or between a hyper cell and a combined cell.
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	In TDD, intra-base-station UL CoMP can be performed between TRPs in a combined cell and between a combined cell and a common cell; but it cannot be performed between combined cells.

- Functions related to energy saving

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	Intra-base-station UL CoMP is not performed on UEs that use bandwidth part 2 (BWP2) for energy saving.
Low-frequency TDD	PDCCH skipping	**PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter	*UE Power Saving*	When intra-base-station UL CoMP and PDCCH skipping are both enabled, PDCCH skipping does not take effect on UL CoMP UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSSG switching	**UE_PWR_SAVING_ENHANCE_SW** and **SSSG_SWITCHING_SW** options of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*UE Power Saving*	UEs for which intra-base-station UL CoMP has taken effect will not switch from a dense search space set group (SSSG) to a sparse one. For intra-base-station UL CoMP to take effect for a UE for which a sparse SSSG is in effect, the UE must switch to a dense SSSG.
Low-frequency TDD	Time-domain power saving BWP (low-frequency TDD)	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	UEs with CoMP in effect will not be switched to BWP2. When a UE is working on BWP2 and the conditions for CoMP to take effect are met, the UE is first switched to BWP1 and then CoMP takes effect.

- Functions related to RAN services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink slot aggregation	**VONR_UL_SLOT_AGGREGATION_SW** and **VOICE_R16_UL_SLOT_AGG_SW** options of the **NRDUCellUlSch**.***VoiceUlSchSwitch*** parameter	*VoNR*	Intra-base-station UL CoMP does not take effect for a VoNR UE in the uplink slot aggregation state. When a VoNR UE for which intra-base-station UL CoMP has taken effect enters the uplink slot aggregation state, the UE will exit intra-base-station UL CoMP.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic uplink slot aggregation	**REDCAP_UL_COVERAGE_ENH_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	For an NR TDD cell, intra-base-station UL CoMP does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect.
Low-frequency TDD	RedCap SSSG switching	• **REDCAP_SSSG_SWITCHING_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter • **REDCAP_SSSG_SWITCH_EXT_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*RedCap*	UEs for which intra-base-station UL CoMP has taken effect will not switch from a dense search space set group (SSSG) to a sparse one. For intra-base-station UL CoMP to take effect for a UE for which a sparse SSSG is in effect, the UE must switch to a dense SSSG.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	**SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter	*Positioning*	When both positioning-dedicated SRS transmission in uplink slots and intra-base-station UL CoMP are enabled: If a UE does not support the positioning-dedicated SRS transmission capability, UL CoMP cannot take effect for the UE. If a UE supports positioning-dedicated SRS transmission and "positioning-dedicated SRS transmission in uplink slots" is enabled for the cooperating neighboring cell, UL CoMP can take effect for the UE and the performance gains are not affected. If a UE supports positioning-dedicated SRS transmission and "positioning-dedicated SRS transmission in uplink slots" is not enabled for the cooperating neighboring cell, UL CoMP can take effect for the UE, but the performance gains decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink 4-layer transmission for a single FWA user	**UL_SU_FOUR_LAYER_SW** option of the **NRDUCellUlRank**.***UlRankAlgoSw*** parameter	*FWA*	If uplink 4-layer transmission for a single FWA user has taken effect, intra-base-station UL CoMP does not take effect for the user.

- Other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Network slicing	Multiple parameters are required for network slicing deployment. For details, see *Network Slicing*.	*Network Slicing*	The network slices configured for a cooperating cell must include UEs' activated network slices.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained and uplink slots	**SRS_INTRF_COORD_S_U_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Channel Management*	When SRS interference coordination based on self-contained and uplink slots is enabled together with intra-base-station UL CoMP: • If the UBBPfw is used, only SRS interference coordination based on self-contained and uplink slots takes effect, and UL CoMP does not take effect. • If the UBBPg is used, both UL CoMP and SRS interference coordination based on self-contained and uplink slots can take effect, but the number of UEs for which UL CoMP takes effect may decrease in the cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	If the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected for the serving cell but deselected for the cooperating cell, intra-base-station UL CoMP does not take effect. If this option is deselected for the serving cell or selected for both the serving and cooperating cells, intra-base-station UL CoMP can take effect.
Low-frequency TDD	Cell-level MRO	**MRO_SCENARIO_IDENT_SW** option of the **gNBMro**.***ScenarioIdentifySwitch*** parameter and **INTRA_FREQ_MRO_SW** option of the **NRCellMro**.***MroSwitch*** parameter	*MRO*	If intra-base-station UL CoMP and cell-level intra-frequency MRO are both enabled, event A3 reporting in UL CoMP is affected. As a result, the identification of UEs in overlapping coverage areas in UL CoMP is affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-level MRO	**UE_MRO_SWITCH** option of the **NRCellMro**.***MroSwitch*** parameter	*MRO*	When intra-base-station UL CoMP and UE-level MRO are both enabled and UL CoMP takes effect for ping-pong UEs, event A3 reporting in UL CoMP is affected. As a result, the identification of UEs in overlapping coverage areas in UL CoMP is affected.
Low-frequency TDD	Uplink massive MIMO multi-layer enhancement	**UL_MMIMO_MULTILAYER_ENHANCE_SW** option of the **NRDUCellUlMimo**.***UlHighLayerMuMimoSwitch*** parameter	*MIMO (TDD)*	The proportion of occasions where UL CoMP takes effect may decrease after multi-layer demodulation performance enhancement takes effect in heavy-load overlapping areas where the conditions for UL CoMP to take effect are met.
Low-frequency TDD	Uplink MU grouping and pairing	**UL_MU_GRP_PAIR_SW** option of the **NRDUCellUlMimo**.***UlMuMimoAlgoSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	When intra-base-station UL CoMP takes effect, enabling uplink MU grouping and pairing may decrease the value of the **N.IntragNB.UL.CoMP.SchUser.Sum** counter.
Low-frequency TDD	Uplink channel densified equalization for high-speed UEs	**HS_UL_LOW_COST_DENSE_EQUAL_SW** option of the **NRDUCellHighSpeedMob**.***HighSpeedAlgoOptSw*** parameter	*High Speed Mobility*	In high-speed scenarios, "uplink channel densified equalization for high-speed UEs" does not take effect for UL CoMP UEs in cooperating TRPs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL 256QAM	**NRDUCellAlgoSwitch**.***Ul256QamSwitch*** set to **UL_256QAM_FIXED**	*Modulation Schemes*	For UEs with UL 256QAM in effect, intra-base-station UL CoMP takes effect for them when the **SBC_COMPATIBLE_WITH_256QAM_SW** option of the **NRDUCellComp**.***UlCompAlgoSwitch*** parameter is selected, and does not take effect when the option is deselected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	**SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter	*Positioning*	When both positioning-dedicated SRS transmission in uplink slots and intra-base-station UL CoMP are enabled: If a UE does not support the positioning-dedicated SRS transmission capability, UL CoMP cannot take effect for the UE. If a UE supports positioning-dedicated SRS transmission and "positioning-dedicated SRS transmission in uplink slots" is enabled for the cooperating neighboring cell, UL CoMP can take effect for the UE and the performance gains are not affected. If a UE supports positioning-dedicated SRS transmission and "positioning-dedicated SRS transmission in uplink slots" is not enabled for the cooperating neighboring cell, UL CoMP can take effect for the UE, but the performance gains decrease.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For TDD: For cells working in 32T32R mode or higher, all NR TDD-capable baseband processing units support this function; for cells working in 1T1R, 2T2R, 4T4R, or 8T8R mode, all NR TDD-capable baseband processing units except the UBBPfw support this function.

All NR-capable main control boards support this function.

RF Modules

For TDD: For cells that work in 32T32R mode or higher, all NR TDD-capable RF modules that comply with the eCPRI protocol support this function; for cells that work in 1T1R, 2T2R, 4T4R, or 8T8R mode, all NR TDD-capable RF modules support this function.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-8 describes the parameters used for function activation.

Table 4-8 Parameters used for activation

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	CoMP Switch	**NRDUCellAlgoSwitch**.*CompSwitch*	INTRA_GNB_UL_COMP_SW	To enable intra-base-station UL CoMP, select this option for both the serving and neighboring cells involved in joint processing.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	CoMP Switch	**NRDUCellAlgoSwitch**.***CompSwitch***	INTRA_GNB_OVERLAPPING_MEAS_SW	To observe the throughput changes before and after UL CoMP is enabled, it is recommended that this option be selected one week before intra-base-station UL CoMP is enabled.
Low-frequency TDD	UL CoMP RSRP Offset	**NRCellComp**.***UlCompRsrpOffset***	None	Retain the default value.
Low-frequency TDD	Overlapping RSRP Offset	**NRCellComp**.***OverlappingRsrpOffset***	None	Set this parameter to the value of **NRCellComp**.***UlCompRsrpOffset***.
Low-frequency TDD	Hysteresis	**NRCellComp**.***Hysteresis***	None	Retain the default value.
Low-frequency TDD	Time to Trigger	**NRCellComp**.***TimeToTrigger***	None	Retain the default value.
Low-frequency TDD	CoMP Configuration Policy	**NRDUCellComp**.***CompConfigPol***	UL_2CELL_COMP_BEAR_POL_SW	Select this option if intra-base-station UL CoMP needs to take effect for UEs with specific QCIs.
Low-frequency TDD	Service Differentiation Algorithm Switch	**gNBQciBearer**.***ServiceDiffAlgoSwitch***	COMP_SW	Select this option for a specific QCI if intra-base-station UL CoMP needs to take effect for UEs with this QCI.
Low-frequency TDD	CoMP Optimization Switch	**NRDUCellComp**.***CompOptSwitch***	COOP_CELL_ALLOC_OPT_SW	It is recommended that this option be selected to increase the probability of intra-base-station UL CoMP taking effect.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

To compare the throughput before and after intra-base-station UL CoMP is enabled, you are advised to set the following parameters one week before this function is enabled. This ensures that sufficient statistics about UE throughput in the overlapping area can be collected for comparison purposes.

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-1;
MOD NRCELLCOMP: NrCellId=0, OverlappingRsrpOffset=-20, UlCompRsrpOffset=-20,
TimeToTrigger=320MS, Hysteresis=2;
MOD NRCELLCOMP: NrCellId=1, OverlappingRsrpOffset=-20, UlCompRsrpOffset=-20,
TimeToTrigger=320MS, Hysteresis=2;
```

Activation Command Examples

```
//Enabling intra-base-station UL CoMP
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CompSwitch=INTRA_GNB_UL_COMP_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CompSwitch=INTRA_GNB_UL_COMP_SW-1;
//Enabling optimized cooperating cell selection
MOD NRDUCELLCOMP: NrDuCellId=0, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-1;
//Setting the CoMP configuration policy
MOD NRDUCELLCOMP: NrDuCellId=0, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-1;
//Selecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling optimized cooperating cell selection
MOD NRDUCELLCOMP: NrDuCellId=0, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-0;
MOD NRDUCELLCOMP: NrDuCellId=1, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-0;
//Deselecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-0;
//Restoring the CoMP configuration policy
MOD NRDUCELLCOMP: NrDuCellId=0, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-0;
MOD NRDUCELLCOMP: NrDuCellId=1, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-0;
//Disabling intra-base-station UL CoMP
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CompSwitch=INTRA_GNB_UL_COMP_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CompSwitch=INTRA_GNB_UL_COMP_SW-0;
//Turning off the switches for intra-base-station overlapping area measurement to reduce signaling
exchange over the air interface (recommended when UE throughput observation in the overlapping area is
not required)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

- Using counters

  Check the **N.IntragNB.UL.CoMP.SchUser.Sum** counter value. If the value is greater than 0, UL CoMP has taken effect.
- Using monitoring items

  On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management** > **NR** > **Cell Performance Monitoring** > **UL CoMP Statistic(Cell) Monitoring**. Check the values of **UL CoMP Combine Users Num** and **UL CoMP Combine RB Num** for the serving cell. If neither of the values is 0, UL CoMP has taken effect.

4.4.3 Network Monitoring

You can evaluate the gains of UL CoMP by comparing the values of the following indicators before and after this function is enabled.

- Uplink throughput of UEs in the overlapping area excluding small-packet throughput, which will increase

  Uplink throughput of UEs in the overlapping area excluding small-packet throughput (Mbit/s) = (**N.ThpVol.UL.CellOverlap** – **N.ThpVol.UL.SmallPkt.CellOverlap**)/ **N.ThpTime.UL.RmvSmallPkt.CellOverlap** x 1000

  Table 4-9 lists the related performance counters.
- **User Uplink Average Throughput (DU)**, which will not decrease

Table 4-9 Uplink performance counters for UEs in the overlapping area

Counter ID	Counter Name
1911820692	**N.ThpVol.UL.CellOverlap**
1911820693	**N.ThpVol.UL.SmallPkt.CellOverlap**
1911820696	**N.ThpTime.UL.RmvSmallPkt.CellOverlap**

NOTICE

Throughput-related performance indicators for UEs in overlapping areas are measured only when the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter is selected. In addition, the **NRCellComp**.***OverlappingRsrpOffset*** parameter needs to be configured, and its value must be the same as the value of either **NRCellComp**.***UlCompRsrpOffset*** or **NRCellComp**.***DlCompRsrpOffset***.

When intra-base-station UL CoMP is enabled, it is recommended that the **NRCellComp**.***OverlappingRsrpOffset*** parameter be set to the same value as the **NRCellComp**.***UlCompRsrpOffset*** parameter.

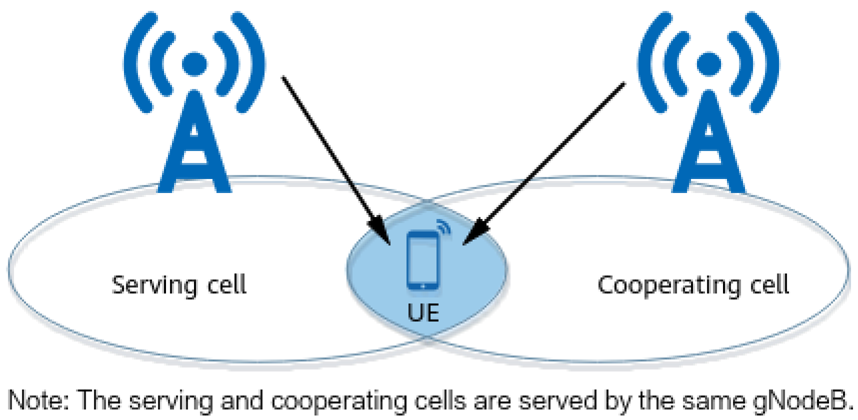
To compare the throughput before and after intra-base-station UL CoMP is enabled, you are advised to perform the preceding configuration one week before intra-base-station UL CoMP is enabled. This ensures that sufficient statistics can be collected for comparison. For detailed operations, see **4.4.1.2 Using MML Commands**.

5 Intra-Base-Station DL CoMP

5.1 Principles

Intra-base-station DL CoMP allows a gNodeB to use the antennas of the serving cell and intra-base-station intra-frequency neighboring cells to jointly transmit PDSCH signals to a UE in the overlapping area, as shown in Figure 5-1. This function increases UE throughput. In this document, intra-base-station DL CoMP is also called intra-base-station joint transmission.

Figure 5-1 Intra-base-station DL CoMP



This function is controlled by the **INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter. For each UE, its serving cell can work with a maximum of two cooperating cells for intra-base-station DL CoMP.

Intra-base-station DL CoMP involves scenarios listed in **Table 5-1**.

Table 5-1 Scenarios involved in intra-base-station DL CoMP

Scenario	Description	Supported RAT	Chapter/ Section
By common cells	Both the serving and cooperating cells of a UE are common cells.	Low-frequency TDD	**5.1.1 Joint Transmission by Common Cells**
By TRPs in a hyper cell	A UE is served by a TRP in a hyper cell, and the gNodeB selects another TRP in the cell for joint data transmission.	Low-frequency TDD	**5.1.2 Joint Transmission by TRPs in a Hyper Cell**
By a combined cell in joint transmission mode and a common cell	A UE is served by a combined cell in joint transmission mode, and the gNodeB selects an intra-base-station common cell as a cooperating cell for joint data transmission.	Low-frequency TDD	**5.1.3 Joint Transmission by a Combined Cell in Joint Transmission Mode and a Common Cell**
By distributed massive MIMO cells	Both the serving and cooperating cells of a UE are LampSite distributed massive MIMO cells.	Low-frequency TDD	**5.1.4 Joint Transmission by Distributed Massive MIMO Cells (NR TDD)**

NOTE

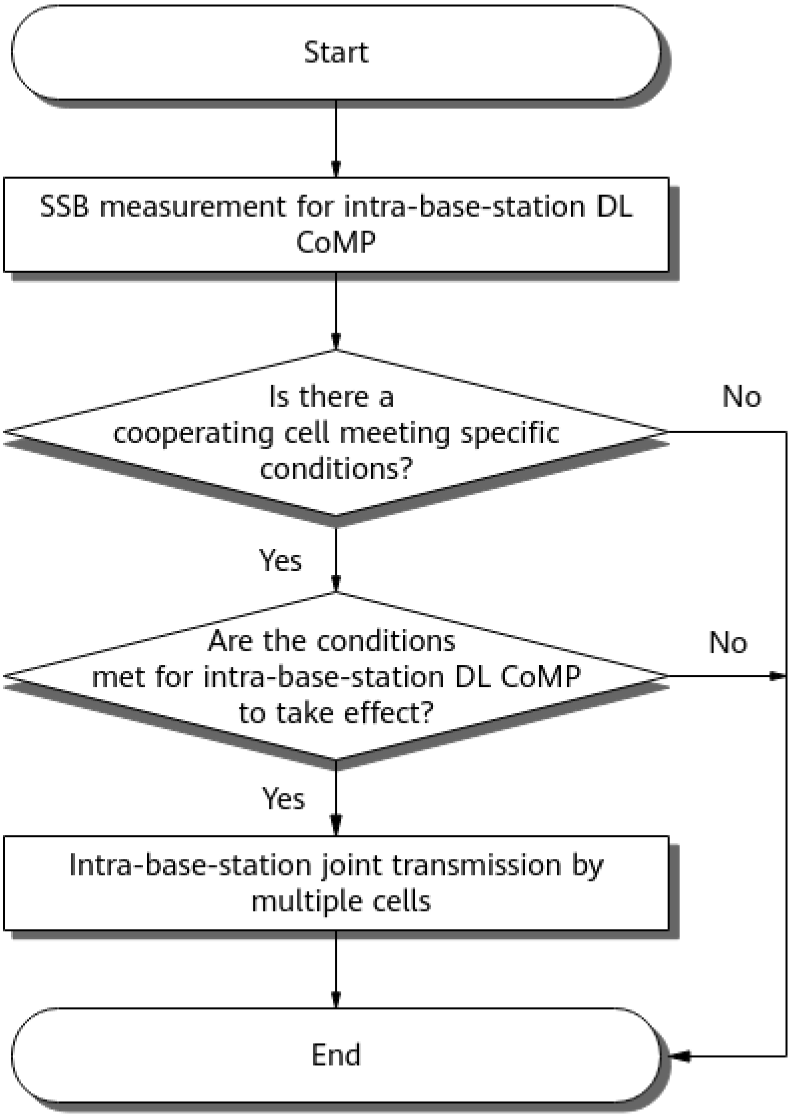
- Common cell: **NRDUCell**.***NrDuCellNetworkingMode*** is set to **NORMAL_CELL**.
- Hyper cell: **NRDUCell**.***NrDuCellNetworkingMode*** is set to **HYPER_CELL**.
- Combined cell in joint transmission mode: **NRDUCell**.***NrDuCellNetworkingMode*** is set to **HYPER_CELL_COMBINE_MODE**, and **NRDUCellMultiTrp**.***DataTransMode*** is set to **JOINT_MODE**.
- LampSite distributed massive MIMO cell: The **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter is selected.
- For details about common cells and TRPs, see *Cell Management*.
- For details about hyper cells, see *Hyper Cell*.
- For details about combined cells in joint transmission mode, see *Cell Combination*.
- For details about distributed massive MIMO cells, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

In addition, the gNodeB support enhancements regarding joint measurement, power, and other aspects to increase the average downlink throughput of intra-base-station DL CoMP UEs. For details, see **5.1.5 Enhanced Functions**.

5.1.1 Joint Transmission by Common Cells

Figure 5-2 shows the processing procedure when a UE is served by a common cell.

Figure 5-2 Joint transmission by common cells



1. Performing SSB measurement for intra-base-station DL CoMP

   The UE starts SSB RSRP measurement after setting up a service and sends an event A3 measurement report when a certain condition is met.

   Off for event A3 in DL CoMP is specified by **NRCellComp**.***DlCompRsrpOffset***. Other information is the same as that for UL CoMP. For details, see **4.1.1 Joint Reception by Common Cells**.

2. Determining whether any cooperating cell meets specific conditions

   The gNodeB selects intra-base-station neighboring cells that meet the following conditions from the event A3 measurement report:

   – Intra-base-station DL CoMP is enabled in the cells.

   – The cells are configured as intra-frequency neighboring cells of the serving cell.

- The PRB usage of the cells is less than the value of **NRDUCellComp**.***DlCompCoCellPrbThld***.
- The number of available downlink RBs for the cells is the same as that for the serving cell.
- In TDD, the TX/RX modes of the cells meet the requirements described in **Table 5-2**.

Table 5-2 TX/RX modes of the serving and cooperating cells required by intra-base-station DL CoMP in TDD common cells

Serving Cell	Cooperating Cell
2T2R	2T2R, 8T8R, 32T32R, or 64T64R
4T4R	4T4R, 8T8R, 32T32R, or 64T64R
8T8R	8T8R, 32T32R, or 64T64R
32T32R or 64T64R	32T32R or 64T64R

- The parameters listed in **Table 5-3** of the cells have the same settings as those of the serving cell.

Table 5-3 Parameters whose settings must be the same in the serving and cooperating cells

Parameter Name	Parameter	Description
Frequency Band	**NRDUCell**.***FrequencyBand***	None
Duplex Mode	**NRDUCell**.***DuplexMode***	None
Downlink NARFCN	**NRDUCell**.***DlNarfcn***	None
Downlink Bandwidth	**NRDUCell**.***DlBandwidth***	None
Subcarrier Spacing	**NRDUCell**.***SubcarrierSpacing***	None
Slot Assignment	**NRDUCell**.***SlotAssignment***	Applicable only to TDD
Cyclic Prefix Length	**NRDUCell**.***CyclicPrefixLength***	None
Downlink DMRS Configuration Type	**NRDUCellPdsch**.***DlDmrsConfigType***	None
Downlink DMRS Maximum Length	**NRDUCellPdsch**.***DlDmrsMaxLength***	None
Downlink Additional DMRS Position	**NRDUCellPdsch**.***DlAdditionalDmrsPos***	None
TRS Period	**NRDUCellCsirs**.***TrsPeriod***	None

Parameter Name	Parameter	Description
Coordination TRS Resource Switch	**JT_TRS_RES_SW** option of the **NRDUCellAlgoSwitch**.***CoordTrsResourceSwitch*** parameter	None
DL CoMP Algorithm Switch	**CSIRS_SCH_ENH_SW** option of the **NRDUCellComp**.***DlCompAlgoSwitch*** parameter	None
Multi-Cell MIMO Switch[a]	**INTER_CLUSTER_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	Applicable only to TDD
Max CSI-RS Power Offset	**NRDUCellChnPwr**.***CsirsMaxPwrOffset***	None
a: The settings of the **INTER_CLUSTER_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter must be identical between the serving and cooperating cells only when the **CSIRS_SCH_ENH_SW** option of the **NRDUCellComp**.***DlCompAlgoSwitch*** parameter is deselected.		

If one or more neighboring cells meet all the preceding conditions, the neighboring cell with the optimal signal quality is treated as a cooperating cell for intra-base-station DL CoMP and the UE that sends the event A3 measurement report is treated as an intra-base-station DL CoMP UE. Otherwise, the procedure ends.

In scenarios with dense deployment of multiple cells, intra-base-station DL CoMP may not take effect due to failure to select a suitable cooperating cell. To address this issue, you are advised to enable optimized cooperating cell selection (by selecting the **COOP_CELL_ALLOC_OPT_SW** option of the **NRDUCellComp**.***CompOptSwitch*** parameter) to expand the range of cells that can be selected as cooperating cells in overlapping areas, thereby increasing the probability of intra-base-station DL CoMP taking effect.

3. Determining whether the conditions for intra-base-station DL CoMP to take effect are met

   a. Performing SRS and CSI-RS measurements for intra-base-station DL CoMP

      To obtain SRS and CSI-RS measurement results for determination, the gNodeB instructs the identified intra-base-station DL CoMP UE to perform the following measurements:

      - Separate SRS measurements in the serving and cooperating cells. If the cooperating cell fails to offer SRS resources in the same time-frequency positions as those allocated in the serving cell of the UE, intra-base-station DL CoMP cannot take effect.

- Joint CSI-RS measurement on the antenna ports of the serving and cooperating cells. CSI-RS refers to CSI-RS for channel measurement (CSI-RS for CM). For details, see *Channel Management*.

b. Determining whether intra-base-station DL CoMP can take effect

Based on the UE-reported SRS and CSI-RS measurement results, the gNodeB checks whether the ratio of the spectral efficiency of joint measurement to that of single-cell measurement is greater than the value of **NRDUCellComp**.***JtSpctEfficiencyThld***. If it is, joint transmission by multiple cells starts. Otherwise, intra-base-station DL CoMP cannot take effect.

4. Performing intra-base-station joint transmission by multiple cells

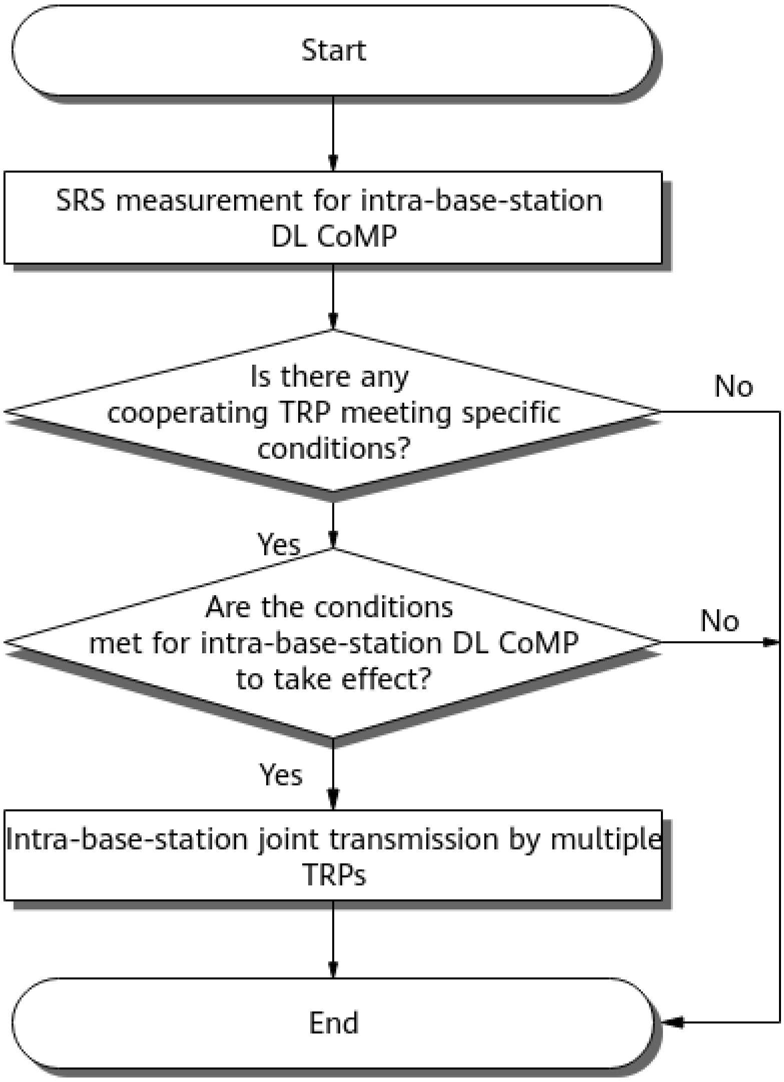
The serving and cooperating cells served by the same base station jointly transmit data to the UE, improving the downlink performance of the UE.

After the UE enters the joint transmission mode, a switchover from joint transmission to single-cell transmission is triggered if the ratio of the spectral efficiency of joint measurement to that of single-cell measurement is less than the **NRDUCellComp**.***JtQuickRollbackSpctEffThld*** value. After the UE enters the single-cell transmission mode, a quick switchover to joint transmission is triggered if the ratio of the spectral efficiency of single-cell measurement to that of joint measurement is less than the **NRDUCellComp**.***JtQuickRollbackSpctEffThld*** value.

5.1.2 Joint Transmission by TRPs in a Hyper Cell

Figure 5-3 shows the processing procedure when a UE is served by a hyper cell.

Figure 5-3 Joint transmission by TRPs in a hyper cell



1. Performing SRS measurement for intra-base-station DL CoMP

   The gNodeB measures the UE SRS RSRP at each TRP in the hyper cell.

2. Determining whether any cooperating TRP meets specific conditions

   If some TRPs meet all of the following conditions, the TRP with the optimal signal quality is treated as a cooperating TRP for intra-base-station DL CoMP.

   – Low-speed scenarios (**NRDUCell**.***HighSpeedFlag*** set to **LOW_SPEED**)

     ■ In TDD, the TX/RX modes of the TRPs meet the requirements described in **Table 5-4**.

Table 5-4 TX/RX modes of the serving and cooperating TRPs required by intra-base-station DL CoMP in low-speed TDD hyper cells

Serving TRP	Cooperating TRP
1T1R	1T1R
2T2R	2T2R or 8T8R

Serving TRP	Cooperating TRP
4T4R	4T4R or 8T8R
8T8R	8T8R
32T32R or 64T64R	32T32R or 64T64R

- High-speed scenarios (**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**)
  - In TDD, the TX/RX modes of the TRPs meet the requirements described in **Table 5-5**.

Table 5-5 TX/RX modes of the serving and cooperating TRPs required by intra-base-station DL CoMP in high-speed TDD hyper cells

Serving TRP	Cooperating TRP
1T1R	1T1R
2T2R	2T2R or 8T8R
4T4R	4T4R or 8T8R
8T8R	8T8R

- The signal quality meets the following requirements: Cooperating TRP RSRP – **NRCellComp**.***Hysteresis*** – Serving TRP RSRP > **NRCellComp**.***DlCompRsrpOffset***.
- The PRB usage of the TRPs is less than the value of **NRDUCellComp**.***DlCompCoCellPrbThld***.

If no TRP meets all the preceding conditions, the procedure ends.

3. Determining whether the conditions for intra-base-station DL CoMP to take effect are met

   a. Performing SRS and CSI-RS measurements for intra-base-station DL CoMP

   To obtain SRS and CSI-RS measurement results for determination, the gNodeB instructs the identified intra-base-station DL CoMP UE to perform the following measurements:

   - Separate SRS measurements in the serving and cooperating TRPs. If a cooperating TRP fails to offer the SRS resources corresponding to those allocated in the serving TRP of the UE, intra-base-station DL CoMP cannot take effect.
   - Joint CSI-RS measurement on the antenna ports of the serving and cooperating TRPs. CSI-RS refers to CSI-RS for channel measurement (CSI-RS for CM). For details, see *Channel Management*.

   b. Determining whether intra-base-station DL CoMP can take effect

   Based on the UE-reported SRS and CSI-RS measurement results, the gNodeB checks whether the ratio of the spectral efficiency of joint measurement to that of single-TRP measurement is greater than the value of **NRDUCellComp**.***JtSpctEfficiencyThld***. If it is, joint transmission by multiple TRPs starts. Otherwise, intra-base-station DL CoMP cannot take effect

4. Performing intra-base-station joint transmission by multiple TRPs

   The serving and cooperating TRPs served by the same base station jointly transmit data to the UE, improving the downlink performance of the UE.

   After the UE enters the joint transmission mode, a switchover from joint transmission to single-TRP transmission is triggered if the ratio of the spectral efficiency of joint measurement to that of single-TRP measurement is less than the **NRDUCellComp**.***JtQuickRollbackSpctEffThld*** value.

5.1.3 Joint Transmission by a Combined Cell in Joint Transmission Mode and a Common Cell

In low-speed scenarios, when a UE is served by a combined cell in joint transmission mode, an intra-base-station common cell can be selected as a cooperating cell for joint transmission. The processing procedure is the same as that for joint transmission by common cells. For details, see **5.1.1 Joint Transmission by Common Cells**.

In TDD, all TRPs in a combined cell in joint transmission mode must work in 2T2R mode, and the cooperating cell must work in 32T32R or 64T64R mode.

5.1.4 Joint Transmission by Distributed Massive MIMO Cells (NR TDD)

When a UE is served by a LampSite distributed massive MIMO cell, the gNodeB can select an intra-base-station distributed massive MIMO cell as a cooperating cell for joint transmission. The processing procedure is the same as that for joint transmission by common cells. For details, see **5.1.1 Joint Transmission by Common Cells**.

If the cells are activated, all TRPs in the distributed massive MIMO cells must work in 4T4R mode.

5.1.5 Enhanced Functions

Intra-base-station DL CoMP can be enhanced in such aspects as joint measurement and power to increase the average downlink throughput of intra-base-station DL CoMP UEs. **Table 5-6** describes the enhancements.

Table 5-6 Enhancements to intra-base-station DL CoMP

Function	Description	Parameter	RAT
Multi-TRS measurement	The UE performs timing and demodulation based on the TRSs sent by the serving and cooperating cells to precisely match the channel conditions of the serving and cooperating cells, improving UE throughput.	**NRDUCellAlgoSwitch**.***MultiTrsMeasSwitch***	Low-frequency TDD

Function	Description	Parameter	RAT
DL CoMP CSI-RS power compensation	CSI-RS power compensation can be performed for an intra-base-station joint transmission UE in periodic time-division CSI-RS measurement scenarios (with **NRDUCellComp**.***JtCsirsTxMode*** set to **PERIODIC_TD_MODE_ABN** or **PERIODIC_TD_MODE_ALL**). This function compensates for the halved CSI-RS power due to separated transmit ports for CSI-RS.	**DL_COMP_CSIRS_PWR_CMP_SW** option of the **NRDUCellChnPwr**.***ChnPwrAlgoSwitch*** parameter	Low-frequency TDD
RMSI enhancement for joint transmission	The remaining minimum system information (RMSI) RB resources can be dynamically adjusted based on actual conditions to increase UE throughput. In TDD, this function requires that the cell bandwidth be lower than 70 MHz or the slot configuration of the cell be dual-period 8:2 (that is, **NRDUCell**.***SlotAssignment*** is set to **8_2_DDDSUUDDDD**).	**JT_RMSI_ENH_SW** option of the **NRDUCellComp**.***DlCompAlgoSwitch*** parameter	Low-frequency TDD
Joint transmission delay compensation	Delay compensation is performed to increase UE throughput when the absolute value of the measured TA difference between the serving and the cooperating cells is greater than or equal to the value of **NRDUCellComp**.***JtDelayCompensationThld***.	**NRDUCellComp**.***JtDelayCompensationThld***	Low-frequency TDD
Joint transmission evaluation optimization	This function enables more accurate evaluation on whether to use single-cell transmission or joint transmission, increasing UE throughput.	**JT_JUDGE_OPT_SW** option of the **NRDUCellComp**.***DlCompAlgoSwitch*** parameter	Low-frequency TDD

Function	Description	Parameter	RAT
Multipoint joint transmission mode	When there are two cooperating TRPs, the serving TRP and the two cooperating TRPs can simultaneously transmit data to an intra-base-station joint transmission UE, improving UE throughput. In TDD, this function is supported only in high-speed hyper cells. This function requires that **NRDUCellComp**.***DlCompEnhancedTransmitMode*** be set to **COMPLEXING_ENH_MODE**.	**NRDUCellComp**.***JtMultiPointTransmitMode*** set to **COV_MULTIPLEX_ENH_HYBRID_MODE**.	Low-frequency TDD
CSI-RS measurement optimization for joint transmission	The UE-level CSI-RS beam type can be specified by **NRDUCellComp**.***UserCsirsBeamType*** to optimize CSI-RS measurement, which improves CSI-RS measurement accuracy and UE throughput.	**JT_CSI_MEAS_OPT_SW** option of the **NRDUCellComp**.***DlCompAlgoSwitch*** parameter	Low-frequency TDD
SRS-based weight enhancement for joint transmission	SRS-based weight enhancement for joint transmission mitigates inter-layer interference between the serving and cooperating cells and increases the signal to interference plus noise ratio (SINR), thereby increasing UE throughput. This function applies only to low-frequency TDD common cells with at least 4T4R, not hyper cells.	**JT_SRS_WEIGHT_ENH_SW** option of the **NRDUCellComp**.***DlCompAlgoSwitch*** parameter	Low-frequency TDD
Rank adaptation for joint transmission	For data transmission for an antenna selection UE, the gNodeB can select the number of layers that better match the signal quality based on the UE-reported rank indication (RI).	**JT_RANK_ADAPT_SW** option of the **NRDUCellComp**.***DlCompAlgoSwitch*** parameter	Low-frequency TDD

Function	Description	Parameter	RAT
Joint transmission power optimization decision	When the SRS RSRP difference between the serving and cooperating cells of a joint transmission UE is greater than the value of **NRDUCellComp**.***JtSrsRsrpDiffThld***, the power of a cell with strong signal strength is reduced by an amount specified by **NRDUCellComp**.***JtPwrBackoffOffset*** to reduce inter-layer interference and increase throughput for the UE.	**NRDUCellComp**.***JtSrsRsrpDiffThld***	Low-frequency TDD
Three-cell coordination for joint transmission	The serving cell and two cooperating cells can simultaneously send data to a joint transmission UE, improving UE throughput. This function applies only to common cells, not hyper cells.	**JT_THREE_CELL_COLLABORATION_SW** option of the **NRDUCellComp**.***DlCompAlgoSwitch*** parameter	Low-frequency TDD
Coordination between cells with different numbers PDCCH symbols	Intra-base-station joint transmission can take effect even when the number of PDCCH symbols in the serving cell is greater than that in the cooperating cell. This function requires that the **PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter be deselected.	**PDCCH_SYM_NUM_DIFF_COORD_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter	Low-frequency TDD

5.2 Network Analysis

5.2.1 Benefits

In intra-frequency continuous networking, intra-base-station DL CoMP increases the receive power and the number of receive paths for UEs in the overlapping area between intra-base-station cells. This increases the MCS index, number of layers, and downlink throughput of these UEs. This function takes effect only for UEs performing full-buffer services. A UE with lower spectral efficiency (indicated by a smaller MCS index or fewer layers) in data transmission using a single cell can benefit more from joint transmission.

For details about how to evaluate the gains brought by this function, see **5.4.3 Network Monitoring**.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

After intra-base-station DL CoMP is enabled, the number of REs used for CSI-RS measurement increases due to joint measurement, and the number of REs available for the PDSCH decreases. In this case, if joint transmission is not used, the peak throughput decreases slightly; if joint transmission is used, the UE throughput increases but the downlink PRB usage slightly decreases.

When intra-base-station DL CoMP and other functions that require aperiodic CSI-RS measurement are enabled at the same time, the number of RRC connection reconfiguration messages increases if the UE supports only one set of resources for aperiodic CSI-RS measurement.

When intra-base-station DL CoMP takes effect during a handover, the following changes may be observed in drive tests: increased handover delay, decreased SSB RSRP, and fluctuation in the values of indicators such as the block error rate and throughput in both the uplink and downlink.

5.3 Requirements

5.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
Low-frequency TDD	FOFD-030204	CoMP	NR0S000CMP00	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRS rate matching	**TRS_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	Enabling intra-base-station joint transmission affects the use of TRS resources. To prevent this impact, the gNodeB reserves TRS resources for cells after TRS rate matching is enabled. For TDD, TRS rate matching or narrowband TRS rate matching must be enabled before intra-base-station joint transmission is enabled.
Low-frequency TDD	Narrowband TRS rate matching	**NARROW_BAND_TRS_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	None	Enabling intra-base-station joint transmission affects the use of TRS resources. To prevent this impact, the gNodeB reserves TRS resources for cells after TRS rate matching is enabled. For TDD, TRS rate matching or narrowband TRS rate matching must be enabled before intra-base-station joint transmission is enabled.
Low-frequency TDD	Slot configuration	**NRDUCell**.***SlotAssignment***	*Standards Compliance*	When intra-base-station DL CoMP is enabled, the **NRDUCell**.***SlotAssignment*** parameter can only be set to **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained slots	**SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Channel Management*	In low-frequency NR TDD scenarios, optimized cooperating cell selection (controlled by the **COOP_CELL_ALLOC_OPT_SW** option of the **NRDUCellComp**.***CompOptSwitch*** parameter) requires that SRS interference coordination based on self-contained slots be enabled first.

5.3.2.2 Mutually Exclusive Functions

- Functions related to cell types

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	VMIMO in DAS-based indoor coverage	**NRDUCellAlgoSwitch**.***VmimoSwitch***	*Indoor Coverage*	Intra-base-station DL CoMP cannot be enabled together with VMIMO in DAS-based indoor coverage.
Low-frequency TDD	Distributed massive MIMO	**DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter	*Distributed Antenna Solutions (Low-Frequency TDD)*	Intra-base-station DL CoMP cannot be enabled in distributed massive MIMO cells served by macro base stations.
Low-frequency TDD	Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	Intra-base-station DL CoMP cannot be enabled in fusion cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Intra-base-station DL CoMP cannot be enabled in combined cells.

- Positioning-related functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	U-TDOA-based positioning/TOA fingerprint-based positioning	**UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	Intra-base-station DL CoMP cannot be enabled together with U-TDOA-based positioning/TOA fingerprint-based positioning.
Low-frequency TDD	SRS field strength-based positioning	**SRS_RSRP_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	Intra-base-station DL CoMP cannot be enabled together with SRS field strength-based positioning.

- Other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRS rate matching optimization	**TRS_RATEMATCH_OPT_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	Intra-base-station DL CoMP cannot be enabled together with TRS rate matching optimization.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Tunnel-based channel imbalance optimization	**CSIRS_BEAM_POWER_BALANCE_SW** option of the **NRDUCellAlgoSwitch**.***BeamOptSwitch*** parameter	*Indoor Coverage*	Intra-base-station DL CoMP does not work with tunnel-based channel imbalance optimization.

5.3.2.3 Function Impacts

- Basic functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUCCH RB adaptation	**PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter	*Channel Management*	When intra-base-station DL CoMP takes effect, it is recommended that PUCCH RB adaptation and CSI reporting period adaptation be enabled. Enabling these functions protects UEs not involved in intra-base-station DL CoMP against service drops that occur due to resource preemption by intra-base-station DL CoMP UEs.
Low-frequency TDD	CSI reporting period adaptation	**CSI_REPORT_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch**.***CsiResoureAlgoSwitch*** parameter	*Channel Management*	When intra-base-station DL CoMP takes effect, it is recommended that PUCCH RB adaptation and CSI reporting period adaptation be enabled. Enabling these functions protects UEs not involved in intra-base-station DL CoMP against service drops that occur due to resource preemption by intra-base-station DL CoMP UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH symbol number adaptation	**UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter	*Channel Management*	• Intra-base-station DL CoMP does not take effect when PDCCH symbol number adaptation is enabled and the number of symbols occupied by the PDCCH in the serving cell is different from that in the cooperating cell. • When the number of PDCCH symbols in the serving cell is greater than that in the cooperating cell, you are advised to select the **PDCCH_SYM_NUM_DIFF_COORD_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter. In this case, intra-base-station DL CoMP can take effect.
Low-frequency TDD	Full-buffer UE-specific CSI-RS for CM	**NRDUCellCsirs**.***CsiUserPeriod***	*Channel Management*	Intra-base-station DL CoMP takes effect only when **NRDUCellComp**.***JtCsirsTxMode*** is set to **PERIODIC_TD_MODE_ABN** or **PERIODIC_TD_MODE_ALL** and **NRDUCellCsirs**.***CsiUserPeriod*** is set to **SLOT20**, **SLOT40**, or **SLOT80**.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	**PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	When rate-matching-pattern-configuration-based PDCCH rate matching and intra-base-station DL CoMP are both enabled: if the **JT_PDCCH_RM_COORD_SW** option of the **NRDUCellComp**.***DlCompAlgoSwitch*** parameter is selected, the functions can take effect simultaneously; if the option is deselected, only one function (which is adaptively determined) can take effect in the overlapping area.
Low-frequency TDD	Rate-matching-pattern-configuration-free PDCCH rate matching	**PDCCH_NO_PATTERN_RATE_MATCH_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter	*Downlink Scheduling*	If the **JT_PDCCH_NO_PATT_RM_COORD_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter is selected, rate-matching-pattern-configuration-free PDCCH rate matching and intra-base-station DL CoMP can take effect simultaneously.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSB rate matching	**SSB_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	Intra-base-station DL CoMP takes effect at a lower probability if the following functions are enabled for both the serving and cooperating cells that use different SSB patterns or PDCCH rate matching patterns: SSB rate matching, rate-matching-pattern-configuration-based PDCCH rate matching, and collaboration between joint transmission and PDCCH rate matching. To increase this probability, it is recommended that the **JT_SSB_RM_COORD_SW** option of the **NRDUCellComp**.***DlCompAlgoExtSwitch*** parameter be selected to enable coordination between joint transmission and SSB rate matching.
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	**PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	
Low-frequency TDD	Collaboration between joint transmission and PDCCH rate matching	**JT_PDCCH_RM_COORD_SW** option of the **NRDUCellComp**.***DlCompAlgoSwitch*** parameter	None	
Low-frequency TDD	Downlink precise RB estimation	**PRECISE_DL_RB_ESTIMATE_SW** option of the **NRDUCellDlSch**.***DlSchAlgoSwitch*** parameter	*Downlink Scheduling*	Intra-base-station DL CoMP may provide higher benefits when it is enabled together with downlink precise RB estimation.

- Functions related to carrier management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When CA is enabled, intra-base-station DL CoMP cannot be performed for CA UEs in their SCells.
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When CA is enabled, intra-base-station DL CoMP cannot be performed for CA UEs in their SCells.
Low-frequency TDD	Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	When inter-gNodeB CA is enabled, intra-base-station DL CoMP cannot be performed for CA UEs in their SCells.
Low-frequency TDD	Multi-frequency convergent scheduling	**MULTI_FREQ_CONVERGENT_SCH_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*Multi-Frequency Smart Aggregation*	Multi-frequency convergent scheduling delivers fewer benefits when it is enabled together with intra-base-station DL CoMP.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	**CA_PRFM_BOOST_BSD_ON_MD_OPT_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*Multi-Frequency Smart Aggregation*	• CA Performance Boost Based on Multi-dimensional Optimization does not take effect for intra-base-station DL CoMP UEs. • On a given gNodeB, if CA Performance Boost Based on Multi-dimensional Optimization is enabled only for some of the cells for which intra-base-station DL CoMP is enabled, the number of cells in which CA Performance Boost Based on Multi-dimensional Optimization takes effect may be less than the number of cells for which CA Performance Boost Based on Multi-dimensional Optimization is enabled. Therefore, it is recommended that CA Performance Boost Based on Multi-dimensional Optimization be enabled for all cells under the gNodeB for which intra-base-station DL CoMP is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Single DCI	**DL_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*Multi-Frequency Smart Aggregation*	Single DCI does not take effect for a UE if intra-base-station DL CoMP and multi-TRS measurement (controlled by the **NRDUCellAlgoSwitch**.***MultiTrsMeasSwitch*** parameter) are enabled in its PCell and the UE supports multi-TRS measurement.

- Functions related to spectrum management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR spectrum sharing	**LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	None	When LTE TDD and NR spectrum sharing takes effect, the number of available downlink RBs in the NR cell dynamically changes. This may affect the probability that intra-base-station DL CoMP takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Operator-specific limitation on the maximum number of available RBs	**MAX_RB_LIMIT_SW** option of the **NRDUCellAlgoSwitch**.***RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*	• When both operator-specific limitation on the maximum number of available RBs and intra-base-station joint transmission are enabled, intra-base-station joint transmission UEs are subject to the limitation on the maximum number of RBs. • UEs of a secondary operator may not be identified as intra-base-station joint transmission UEs when both operator-specific limitation on the maximum number of available RBs and intra-base-station joint transmission are enabled and the **NRDUCellRes**.***DlRbMaxRatio*** parameter is set to a value smaller than 50% for this operator.

- Functions related to cell networking planning

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Intra-base-station joint transmission is supported only between TRPs in a hyper cell, but not between hyper cells.

- Functions related to energy saving

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	**RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	When RF channel intelligent shutdown takes effect in a cell, intra-base-station joint transmission will stop taking effect for UEs already in such transmission in this cell and will not take effect for UEs newly admitted to this cell. After the cell exits the RF channel intelligent shutdown state, intra-base-station joint transmission takes effect again.
Low-frequency TDD	LNR coordinated channel shutdown	**MULTI_RAT_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Multi-RAT Coordinated Energy Saving*	When LNR coordinated channel shutdown takes effect in a cell, intra-base-station joint transmission will stop taking effect for UEs already in such transmission in this cell and will not take effect for UEs newly admitted to this cell. After the cell exits the LNR coordinated channel shutdown state, intra-base-station joint transmission takes effect again.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	Intra-base-station joint transmission is not performed for UEs that use BWP2 for power saving.
Low-frequency TDD	PDCCH skipping	**PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter	*UE Power Saving*	When intra-base-station DL CoMP and PDCCH skipping are both enabled, PDCCH skipping does not take effect on DL CoMP UEs.
Low-frequency TDD	SSSG switching	**UE_PWR_SAVING_ENHANCE_SW** and **SSSG_SWITCHING_SW** options of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*UE Power Saving*	UEs for which intra-base-station DL CoMP has taken effect will not switch from a dense SSSG to a sparse one. For intra-base-station DL CoMP to take effect for a UE for which a sparse SSSG is in effect, the UE must switch to a dense SSSG.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	• After discontinuous reception (DRX) is enabled, UEs are less likely to be identified as intra-base-station joint transmission UEs. • After DRX is enabled: if the function of DRX exit of UEs performing full-buffer services (controlled by the **FULL_BUFFER_UE_EXIT_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter) is enabled, DRX (regardless of the DRX sleep or active time) does not take effect for a UE identified as an intra-base-station joint transmission UE; if the function of DRX exit of UEs performing full-buffer services is disabled, a UE identified as an intra-base-station joint transmission UE cannot enter the DRX sleep time because the aperiodic CSI-RS for CM is used.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent power control energy saving	**SMART_PWR_CTRL_ENERGY_SAVE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	Power reduction does not take effect for DL CoMP UEs.
Low-frequency TDD	Dynamic RF channel shutdown	**RF_CHN_DYN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter	*Energy Conservation and Emission Reduction*	Dynamic RF channel shutdown does not take effect in a cell serving DL CoMP UEs.
Low-frequency TDD	NR low power consumption mode	**LOW_PWR_MODE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Green Site*	Only low power consumption mode takes effect when it is enabled together with intra-base-station DL CoMP and a customized backup power saving policy is used in low power consumption mode.
Low-frequency TDD	Time-domain power saving BWP (low-frequency TDD)	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	UEs with CoMP in effect will not be switched to BWP2. When a UE is working on BWP2 and the conditions for CoMP to take effect are met, the UE is first switched to BWP1 and then CoMP takes effect.

- Functions related to RAN services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	Intra-base-station DL CoMP does not take effect for RedCap UEs.
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	**SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter	*Positioning*	When both positioning-dedicated SRS transmission in uplink slots and intra-base-station DL CoMP are enabled: If a UE does not support the positioning-dedicated SRS transmission capability, DL CoMP cannot take effect for the UE. If a UE supports positioning-dedicated SRS transmission, DL CoMP can take effect for the UE.

- Functions related to power sharing

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	**NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***DynPowerSharingSwitch*** parameter	*NR Inter-Carrier Dynamic Power Allocation*	NR inter-carrier dynamic power sharing may provide fewer benefits when it is enabled together with intra-base-station DL CoMP.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	**LTE_NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***DynPowerSharingSwitch*** parameter	*LTE and NR Power Allocation*	When intra-base-station DL CoMP and LTE TDD and NR Flash Dynamic Power Sharing are both enabled, the power sharing gains on the NR side may decrease.
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	**LNR_MULTI_BAND_PWR_ADPT_SCH_SW** option of the **NRDUCellAlgoSwitch**.***DynPowerSharingSwitch*** parameter	*LTE and NR Power Allocation*	When LTE and NR multi-band power adaptive scheduling and intra-base-station DL CoMP are both enabled, the benefits of power adaptive scheduling may decrease on the NR side.
Low-frequency TDD	L/NR Inter-band Deep Coordination	**LNR_INTER_BAND_DEEP_COORD_SW** option of the **NRDUCellResShrCfg**.***DynPwrShrAlgoSwitch*** parameter	*LTE and NR Power Allocation*	When L/NR Inter-band Deep Coordination and intra-base-station DL CoMP are both enabled, the power sharing gains on the NR side may decrease.
Low-frequency TDD	NR multi-band power adaptive scheduling	**NR_MULTI_BAND_PWR_ADAPT_SCH_SW** option of the **NRDUCellAlgoSwitch**.***DynPowerSharingSwitch*** parameter	*NR Inter-Carrier Dynamic Power Allocation*	NR multi-band power adaptive scheduling may provide fewer benefits when it is enabled together with intra-base-station DL CoMP.

- Functions related to interference coordination

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS remote interference avoidance	**SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	When SRS remote interference avoidance is enabled in the serving cell but disabled in the cooperating cell, intra-base-station DL CoMP does not take effect.
Low-frequency TDD	Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	Coordinated interference management is not performed for UEs involved in intra-base-station DL CoMP.
Low-frequency TDD	External interference avoidance	**UL_UE_LEVEL_INTRF_AVOID** option of the **NRDUCellIntrfIdent.*IntrfOptimizationMethod*** parameter and **NRDUCellIntrfIdent**.***IntrfIdentificationMethod*** set to **UL_MANUAL** or **UL_AUTO**	*Interference Avoidance*	After external interference avoidance is enabled, there is no SRS transmission in the interference range. As a result, the probability that intra-base-station DL CoMP takes effect decreases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency offset measurement optimization based on interference suppression	**INTRF_FREQOFFSET_OPT_SW** option of the **NRDUCellPusch**.***PuschMeasSw*** parameter	*Interference Avoidance*	After frequency offset measurement optimization based on interference suppression takes effect, the average downlink cell throughput may increase in scenarios where intra-base-station DL CoMP is enabled.
Low-frequency TDD	Interference-based subband robust weights	**INTRF_BASED_ROBUST_WEIGHT_SW** option of the **NRDUCellIntrfIdent**.***ChnMeasCtrlSw*** parameter	*Interference Avoidance*	Interference-based subband robust weights do not take effect for UEs that report event A3 for CoMP.
Low-frequency TDD	Remote interference adaptive avoidance	**RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	Intra-base-station DL CoMP does not take effect in self-contained slots if the following conditions are met: (1) Remote interference adaptive avoidance is enabled. (2) There are downlink symbols in the self-contained slots of the serving and cooperating cells. (3) The number of downlink symbols in each self-contained slot of the serving cell is greater than that in such a slot of the cooperating cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-cell interference avoidance	**INTER_CELL_INTRF_AVOID_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Massive MIMO iBeam (Low-Frequency TDD)*	Inter-cell interference avoidance does not apply to UEs for which intra-base-station DL CoMP is in effect.
Low-frequency TDD	Common channel interference avoidance	**SSB_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter	*Interference Avoidance*	Enabling common channel interference avoidance decreases the probability of intra-base-station DL CoMP taking effect.

- Functions related to MIMO

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CCE resource optimization	**CCE_RESOURCE_OPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter	*MIMO (TDD)*	If CCE resource optimization is enabled, the proportion of occasions where intra-base-station DL CoMP takes effect may decrease. It is recommended that the **PDCCH_SYM_NUM_DIFF_COORD_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter be selected for optimization.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart PDCCH symbol allocation	**PDCCH_SYM_SMART_ALLOC_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	When DL CoMP takes effect, enabling smart PDCCH symbol allocation may decrease the uplink CCE allocation success rate of UEs and the average uplink UE throughput.
Low-frequency TDD	PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	Intra-base-station DL CoMP UEs do not support PDSCH MU-MIMO.
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	**DL_SU_EIGHT_LAYER_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter	*FWA*	If downlink 8-layer transmission for a single FWA user has taken effect, intra-base-station DL CoMP does not take effect for the user.
Low-frequency TDD	Robust weights	**DL_ROBUST_WEIGHT_SW** option of the **NRDUCellPdschPrecode**.***PrecodingAlgoSwitch*** parameter	*Massive MIMO iBeam (Low-Frequency TDD)*	Enabling robust weights may decrease the downlink user-perceived rate of DL CoMP UEs.

- Other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag***	*High Speed Mobility*	When High-speed Railway Superior Experience is enabled, only intra-base-station DL CoMP by TRPs in a hyper cell is supported.
Low-frequency TDD	Network slicing	Multiple parameters are required for network slicing deployment. For details, see *Network Slicing*.	*Network Slicing*	The network slices configured for a cooperating cell must include UEs' activated network slices.
Low-frequency TDD	Cell-level MRO	**MRO_SCENARIO_IDENT_SW** option of the **gNBMro**.***ScenarioIdentifySwitch*** parameter and **INTRA_FREQ_MRO_SW** option of the **NRCellMro**.***MroSwitch*** parameter	*MRO*	If intra-base-station DL CoMP and cell-level intra-frequency MRO are both enabled, event A3 reporting in DL CoMP is affected. As a result, the identification of UEs in overlapping coverage areas in DL CoMP is affected.
Low-frequency TDD	UE-level MRO	**UE_MRO_SWITCH** option of the **NRCellMro**.***MroSwitch*** parameter	*MRO*	When both intra-base-station DL CoMP and UE-level MRO are enabled and DL CoMP takes effect for ping-pong-handover UEs, event A3 reporting in DL CoMP is affected. As a result, the identification of UEs in overlapping coverage areas in DL CoMP is affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink type 1 scheduling for slice groups with the TTI-based control policy	**DL_TYPE1_SCH_SW** option of the **NRDUCellRes**.***SchedulingPolicySwitch*** parameter	*Network Slicing*	When downlink type 1 scheduling for slice groups with the TTI-based control policy takes effect, intra-base-station DL CoMP does not take effect for UEs in the slice groups.
Low-frequency TDD	Uplink channel densified equalization for high-speed UEs	**HS_UL_LOW_COST_DENSE_EQUAL_SW** option of the **NRDUCellHighSpeedMob**.***HighSpeedAlgoOptSw*** parameter	*High Speed Mobility*	In high-speed scenarios, "uplink channel densified equalization for high-speed UEs" does not take effect for DL CoMP UEs in cooperating TRPs.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Intra-base-station DL CoMP between distributed massive MIMO cells requires DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

In TDD:

- Joint transmission by common cells: For cells working in 8T8R mode or higher, all NR TDD-capable baseband processing units support this function. For cells working in 1T1R, 2T2R, or 4T4R mode, all NR TDD-capable baseband processing units except the UBBPfw support this function.
- Joint transmission by TRPs in a hyper cell: If all TRPs adopt the same TX/RX mode or if some TRPs work in 32T32R mode and others work in 64T64R mode, all NR TDD-capable baseband processing units support this function. In

other conditions, all NR TDD-capable baseband processing units except the UBBPfw support this function.

- For details about the board requirements for a LampSite distributed massive MIMO cell, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

All NR-capable main control boards support this function.

RF Modules

In TDD: For cells that work in 32T32R mode or higher, all NR TDD-capable RF modules that comply with the eCPRI protocol support this function; for cells that work in 1T1R, 2T2R, 4T4R, or 8T8R mode, all NR TDD-capable RF modules support this function.

For details about the RF module requirements for a LampSite distributed massive MIMO cell, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

5.3.4 Others

Requirements for the moving speeds of UEs are as follows:

- When the High-speed Railway Superior Experience function is enabled, only intra-base-station joint transmission by TRPs in a hyper cell is supported. In this case, intra-base-station joint transmission has no requirements for the UE moving speed.
- When the High-speed Railway Superior Experience function is disabled, pre-correction of the downlink frequency shift is not supported. In this case, the moving speeds of intra-base-station joint transmission UEs must be less than or equal to 60 km/h to avoid negative impacts of intra-base-station joint transmission.

UEs must support aperiodic CSI-RS measurement. If the value of the *maxNumberAperiodicCSI-PerBWP-ForCSI-Report* IE in **CSI-Report** of the UECapabilityInformation message is not 0, the UE supports this function. If a UE is incompatible with aperiodic CSI-RS measurement, the UE may not work properly when intra-base-station joint transmission is enabled. In this case, contact engineers from Huawei or the UE vendor.

Intra-base-station joint transmission cannot take effect if UEs do not support related CSI-RS capabilities (such as the number of CSI-RS ports, CSI-RS resources, and CSI-RS reporting).

Multi-TRS measurement (controlled by the **NRDUCellAlgoSwitch**.***MultiTrsMeasSwitch*** parameter) requires that the UE can measure multiple sets of TRSs. The UE can measure multiple sets of TRSs if the value of the **maxNumberActiveTCI-PerBWP** field of the corresponding band is greater than or equal to n2 in the UE capability information.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-7 describes the parameters used for function activation.

Table 5-7 Parameters used for activation

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	CoMP Switch	**NRDUCellAlgoSwitch**.***CompSwitch***	INTRA_GNB_DL_JT_SW	To enable intra-base-station joint transmission, select this option for both the serving and neighboring cells involved in joint processing.
Low-frequency TDD	CoMP Switch	**NRDUCellAlgoSwitch**.***CompSwitch***	INTRA_GNB_OVERLAPPING_MEAS_SW	It is recommended that this option be selected one week before intra-base-station joint transmission is enabled to observe the throughput changes before and after intra-base-station joint transmission is enabled.
Low-frequency TDD	DL CoMP RSRP Offset	**NRCellComp**.***DlCompRsrpOffset***	None	Retain the default value.
Low-frequency TDD	Overlapping RSRP Offset	**NRCellComp**.***OverlappingRsrpOffset***	None	Set this parameter to the value of **NRCellComp**.***DlCompRsrpOffset***.
Low-frequency TDD	Hysteresis	**NRCellComp**.***Hysteresis***	None	Retain the default value.
Low-frequency TDD	Time to Trigger	**NRCellComp**.***TimeToTrigger***	None	Retain the default value. This parameter is not involved in Hyper Cell networking scenarios.
Low-frequency TDD	JT Spectral Efficiency Threshold	**NRDUCellComp**.***JtSpctEfficiencyThld***	None	Retain the default value.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	JT Quick Rollback Spectral Efficiency Thld	**NRDUCellComp**.***JtQuickRollbackSpctEffThld***	None	Retain the default value.
Low-frequency TDD	DL CoMP Coordinating Cell PRB Threshold	**NRDUCellComp**.***DlCompCoCellPrbThld***	None	Retain the default value.
Low-frequency TDD	DL CoMP Enhanced Transmit Mode	**NRDUCellComp**.***DlCompEnhancedTransmitMode***	None	The value **COMPLEXING_ENH_MODE** is recommended when intra-base-station DL CoMP is enabled.
Low-frequency TDD	DL CoMP Algorithm Switch	**NRDUCellComp**.***DlCompAlgoSwitch***	JT_RMSI_ENH_SW	It is recommended that this option be selected for a TDD cell when any of the following conditions is met: ● The cell adopts the dual-period 8:2 slot configuration. ● The cell adopts a slot configuration other than dual-period 8:2, and the cell bandwidth is less than 70 MHz.
Low-frequency TDD	DL CoMP Algorithm Switch	**NRDUCellComp**.***DlCompAlgoSwitch***	JT_JUDGE_OPT_SW	It is recommended that this option be selected. This function is supported in both common cell and Hyper Cell networking scenarios.
Low-frequency TDD	JT Multi-Point Transmission Mode	**NRDUCellComp**.***JtMultiPointTransmitMode***	None	The value **COV_MULTIPLEX_ENH_HYBRID_MODE** is recommended only for high-speed hyper cells in the case of TDD operation when intra-base-station DL CoMP is enabled.
Low-frequency TDD	Multi-TRS Measurement Switch	**NRDUCellAlgoSwitch**.***MultiTrsMeasSwitch***	None	It is recommended that this switch be turned on in high-speed scenarios.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	JT Delay Compensation Threshold	**NRDUCellComp**.***JtDelayCompensationThld***	None	Retain the default value. For sites that have been upgraded from a version earlier than V100R005C10, this parameter needs to be manually set to the default value **15**.
Low-frequency TDD	JT CSI-RS Transmit Mode	**NRDUCellComp**.***JtCsirsTxMode***	None	Use the recommended value.
Low-frequency TDD	Downlink PDSCH Algorithm Switch	**NRDUCellPdsch**.***DlPdschAlgoSwitch***	RMSI_RES_RSV_OPT_SW	It is recommended that this option be selected.
Low-frequency TDD	JT Mode Spectral Eff Threshold	**NRDUCellComp**.***JtModeSpecEffThld***	None	For TDD, when **NRDUCellTrp**.***TxRxMode*** is set to **32T32R** or **64T64R**, the value **200** is recommended; when **NRDUCellTrp**.***TxRxMode*** is set to any other value, the value **100** is recommended.
Low-frequency TDD	DL CoMP Algorithm Extension Switch	**NRDUCellComp**.***DlCompAlgoExtSwitch***	JT_SSB_RM_COORD_SW	When SSB rate matching is enabled for both the serving and cooperating cells, it is recommended that this option be selected to increase the probability that intra-base-station DL CoMP takes effect.
Low-frequency TDD	DL CoMP Algorithm Switch	**NRDUCellComp**.***DlCompAlgoSwitch***	JT_SRS_WEIGHT_ENH_SW	It is recommended that this option be selected for the serving and neighboring cells involved in joint processing. Note: In 4T4R and 8T8R scenarios, this function takes effect only when the UBBPg or later series boards are used.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	DL CoMP Algorithm Switch	**NRDUCellComp**.***DlCompAlgoSwitch***	JT_RANK_ADAPT_SW	It is recommended that this option be selected when there are antenna selection UEs on the network. This function is supported in both common cell and Hyper Cell networking scenarios.
Low-frequency TDD	DL CoMP Algorithm Switch	**NRDUCellComp**.***DlCompAlgoSwitch***	JT_CSI_MEAS_OPT_SW	It is recommended that this option be selected for the serving and neighboring cells involved in joint processing. This function is supported only in cells that work in 32T32R mode or higher.
Low-frequency TDD	User CSI-RS Beam Type	**NRDUCellComp**.***UserCsirsBeamType***	None	Use the recommended value.
Low-frequency TDD	JT Power Backoff Offset	**NRDUCellComp**.***JtPwrBackoffOffset***	None	Use the recommended value.
Low-frequency TDD	JT SRS RSRP Difference Threshold	**NRDUCellComp**.***JtSrsRsrpDiffThld***	None	Use the recommended value.
Low-frequency TDD	JT Mode Probe Offset	**NRDUCellComp**.***JtModeProbeOffset***	None	Use the recommended value.
Low-frequency TDD	Single Cell Trans Probe Offset	**NRDUCellComp**.***SingleCellTransProbeOffset***	None	Use the recommended value.
Low-frequency TDD	PDCCH Algorithm Enhancement Switch	**NRDUCellPdcch**.***PdcchAlgoEnhSwitch***	PDCCH_SYM_NUM_DIFF_COORD_SW	It is recommended that this option be selected.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	CoMP Optimization Switch	**NRDUCellComp**.***CompOptSwitch***	COOP_CELL_ALLOC_OPT_SW	It is recommended that this option be selected to increase the probability of intra-base-station DL CoMP taking effect.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

To compare the throughput before and after intra-base-station joint transmission is enabled, you are advised to turn on the following switches one week before intra-base-station joint transmission is enabled. This ensures that sufficient statistics about UE throughput in the overlapping area can be collected for comparison purposes.

```
//Enabling intra-base-station overlapping area measurement
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-1;
//Setting parameters related to intra-base-station overlapping area measurement
MOD NRCELLCOMP: NrCellId=0, OverlappingRsrpOffset=-18, DlCompRsrpOffset=-18,
TimeToTrigger=320MS, Hysteresis=2;
MOD NRCELLCOMP: NrCellId=1, OverlappingRsrpOffset=-18, DlCompRsrpOffset=-18,
TimeToTrigger=320MS, Hysteresis=2;
```

Activation Command Examples

```
//Enabling intra-base-station joint transmission
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CompSwitch=INTRA_GNB_DL_JT_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CompSwitch=INTRA_GNB_DL_JT_SW-1;
//Setting the JtSpctEfficiencyThld parameter
MOD NRDUCELLCOMP: NrDuCellId=0, JtSpctEfficiencyThld=110;
MOD NRDUCELLCOMP: NrDuCellId=1, JtSpctEfficiencyThld=110;
//Setting the JtQuickRollbackSpctEffThld parameter
MOD NRDUCELLCOMP: NrDuCellId=0, JtQuickRollbackSpctEffThld=95;
MOD NRDUCELLCOMP: NrDuCellId=1, JtQuickRollbackSpctEffThld=95;
//Setting the DlCompCoCellPrbThld parameter
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompCoCellPrbThld=30;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompCoCellPrbThld=30;
//Setting the DlCompEnhancedTransmitMode parameter
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompEnhancedTransmitMode=COMPLEXING_ENH_MODE;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompEnhancedTransmitMode=COMPLEXING_ENH_MODE;
//Setting the multipoint joint transmission mode (applicable only to high-speed hyper cells in the case of
TDD operation)
MOD NRDUCELLCOMP: NrDuCellId=0, JtMultiPointTransmitMode=COV_MULTIPLEX_ENH_HYBRID_MODE;
MOD NRDUCELLCOMP: NrDuCellId=1, JtMultiPointTransmitMode=COV_MULTIPLEX_ENH_HYBRID_MODE;
//Enabling RMSI enhancement for joint transmission
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompAlgoSwitch=JT_RMSI_ENH_SW-1;
```



```
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompAlgoSwitch=JT_RMSI_ENH_SW-1;
//Enabling optimization of decision on using single-cell transmission or joint transmission
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompAlgoSwitch=JT_JUDGE_OPT_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompAlgoSwitch=JT_JUDGE_OPT_SW-1;
//Enabling measurement of multiple sets of TRSs (recommended in high-speed scenarios)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MultiTrsMeasSwitch=ON;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, MultiTrsMeasSwitch=ON;
//Setting the JtDelayCompensationThld parameter
MOD NRDUCELLCOMP: NrDuCellId=0, JtDelayCompensationThld=15;
MOD NRDUCELLCOMP: NrDuCellId=1, JtDelayCompensationThld=15;
//Setting the JtCsirsTxMode parameter
MOD NRDUCELLCOMP: NrDuCellId=0, JtCsirsTxMode=APERIODIC_MODE;
MOD NRDUCELLCOMP: NrDuCellId=1, JtCsirsTxMode=APERIODIC_MODE;
//Enabling RMSI resource reservation optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=RMSI_RES_RSV_OPT_SW-1;
MOD NRDUCELLPDSCH: NrDuCellId=1, DlPdschAlgoSwitch=RMSI_RES_RSV_OPT_SW-1;
//Setting the JtModeSpecEffThld parameter (Value 200 is recommended for 32T32R or 64T64R scenarios,
and value 100 is recommended for other scenarios. The following are MML command examples in other
scenarios.)
MOD NRDUCELLCOMP: NrDuCellId=0,JtModeSpecEffThld=100;
MOD NRDUCELLCOMP: NrDuCellId=1,JtModeSpecEffThld=100;
//Enabling coordination between joint transmission and SSB rate matching when SSB rate matching is
enabled for both the serving and cooperating cells
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompAlgoExtSwitch=JT_SSB_RM_COORD_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompAlgoExtSwitch=JT_SSB_RM_COORD_SW-1;
//(Only for TDD) Enabling SRS-based weight enhancement for joint transmission
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompAlgoSwitch=JT_SRS_WEIGHT_ENH_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompAlgoSwitch=JT_SRS_WEIGHT_ENH_SW-1;
//(Only for TDD) Enabling rank adaptation for joint transmission
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompAlgoSwitch=JT_RANK_ADAPT_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompAlgoSwitch=JT_RANK_ADAPT_SW-1;
//(Only for TDD) Enabling CSI-RS measurement optimization for joint transmission
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompAlgoSwitch=JT_CSI_MEAS_OPT_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompAlgoSwitch=JT_CSI_MEAS_OPT_SW-1;
//(Only for TDD) Setting the UserCsirsBeamType parameter
MOD NRDUCELLCOMP: NrDuCellId=0,UserCsirsBeamType=TYPE0;
MOD NRDUCELLCOMP: NrDuCellId=1,UserCsirsBeamType=TYPE0;
//(Only for TDD) Setting the JtPwrBackoffOffset parameter
MOD NRDUCELLCOMP: NrDuCellId=0, JtPwrBackoffOffset=6;
MOD NRDUCELLCOMP: NrDuCellId=1, JtPwrBackoffOffset=6;
//(Only for TDD) Setting the JtSrsRsrpDiffThld parameter
MOD NRDUCELLCOMP: NrDuCellId=0, JtSrsRsrpDiffThld=6;
MOD NRDUCELLCOMP: NrDuCellId=1, JtSrsRsrpDiffThld=6;
//(Only for TDD) Setting the JtModeProbeOffset parameter
MOD NRDUCELLCOMP: NrDuCellId=0,JtModeProbeOffset=100;
MOD NRDUCELLCOMP: NrDuCellId=1,JtModeProbeOffset=100;
//(Only for TDD) Setting the SingleCellTransProbeOffset parameter
MOD NRDUCELLCOMP: NrDuCellId=0, SingleCellTransProbeOffset=100;
MOD NRDUCELLCOMP: NrDuCellId=1, SingleCellTransProbeOffset=100;
//Enabling coordination between serving and cooperating cells with different numbers of PDCCH symbols
(TDD only)
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_SYM_NUM_DIFF_COORD_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoEnhSwitch=PDCCH_SYM_NUM_DIFF_COORD_SW-1;
//Enabling optimized cooperating cell selection
MOD NRDUCELLCOMP: NrDuCellId=0, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling intra-base-station joint transmission
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CompSwitch=INTRA_GNB_DL_JT_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CompSwitch=INTRA_GNB_DL_JT_SW-0;
//Turning off the switches for intra-base-station overlapping area measurement to reduce signaling
exchange over the air interface (recommended when UE throughput observation in the overlapping area is
```



```
not required)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

- Using counters

  For TDD, check the **N.IntragNB.DL.NCJT.SchUser.Sum** counter value. If the value is greater than 0, the intra-base-station joint transmission function has taken effect.
- Using monitoring items

  On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management** > **NR** > **Cell Performance Monitoring** > **DL JT Statistic(Cell) Monitoring**. Check the values of **DL JT Schedule Users Num** and **DL JT Schedule RB Num** for the serving cell. If neither of the values is 0, intra-base-station joint transmission has taken effect.

5.4.3 Network Monitoring

You can evaluate the gains of intra-base-station joint transmission by comparing the values of the following indicators before and after this function is enabled.

- Downlink throughput of UEs in the overlapping area, which will increase

  Downlink throughput of UEs in the overlapping area (Mbit/s) = (**N.ThpVol.DL.CellOverlap** – **N.ThpVol.DL.LastSlot.CellOverlap**)/ **N.ThpTime.DL.RmvLastSlot.CellOverlap** x 1000

  Table 5-8 lists the related performance counters.
- **User Downlink Average Throughput (DU)**, which will not decrease

Table 5-8 Downlink performance counters for UEs in the overlapping area

Counter ID	Counter Name
1911820651	**N.ThpVol.DL.CellOverlap**
1911823024	**N.ThpVol.DL.LastSlot.CellOverlap**
1911823026	**N.ThpTime.DL.RmvLastSlot.CellOverlap**

NOTICE

Throughput-related performance indicators for UEs in overlapping areas are measured only when the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter is selected. In addition, the **NRCellComp**.***OverlappingRsrpOffset*** parameter needs to be configured, and its value must be the same as the value of either **NRCellComp**.***UlCompRsrpOffset*** or **NRCellComp**.***DlCompRsrpOffset***.

When intra-base-station DL CoMP is enabled, it is recommended that the **NRCellComp**.***OverlappingRsrpOffset*** parameter be set to the same value as the **NRCellComp**.***DlCompRsrpOffset*** parameter.

To compare the throughput before and after intra-base-station DL CoMP is enabled, you are advised to perform the preceding configuration one week before intra-base-station DL CoMP is enabled. This ensures that sufficient statistics can be collected for comparison. For detailed operations, see **5.4.1.2 Using MML Commands**.

When the probability for intra-base-station joint transmission to take effect is low due to a small overlapping coverage area between cells, the gains monitored using the preceding indicators are not significant. In this case, the gains after this function is enabled can be monitored by using the single-UE monitoring method. The procedure is detailed as follows: On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management** > **NR** > **User Performance Monitoring** > **User Common Monitoring**, and check the changes of the downlink RLC throughput of UEs.

The single-UE monitoring method can be used to observe the gains provided by this function in the following scenarios:

- Joint transmission between common 32T32R or 64T64R TDD cells
- Joint transmission by TRPs in a hyper cell

In scenarios of joint transmission by 8T8R, 4T4R, or 2T2R common TDD cells, the preceding single-UE monitoring method can be used to observe the gains provided by this function when UEs are stationary.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *Cell Management*
- *Carrier Aggregation*
- *UE Power Saving*
- *Multi-Operator Sharing*
- *Hyper Cell*
- *Cell Combination*
- *High Speed Mobility*
- *Remote Interference Management (Low-Frequency TDD)*
- *Channel Management*
- *Downlink Scheduling*
- *Network Slicing*
- *Standards Compliance*
- *DRX*
- *Energy Conservation and Emission Reduction*
- *Multi-RAT Coordinated Energy Saving*
- *Interference Avoidance*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *MIMO (TDD)*
- *LTE and NR Power Allocation*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *MRO*
- *Positioning*
- *FWA*
- *Multi-Frequency Smart Aggregation*
- *RedCap*
- *Fusion Cell (Low-Frequency TDD)*

- *VoNR*
- *Green Site*
- *NR Inter-Carrier Dynamic Power Allocation*
- *License Control Item Lists*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Indoor Coverage*
- *Modulation Schemes*

